Applying the X-Partition / Red-Blue Pebble Framework to QR and Schur Decompositions

Panel factorization via shifted CholeskyQR3 (Fukaya et al. 2018)

Revised manuscript

Theoretical Algorithm Design Sketch

Contents

Revision Summary

This section consolidates cross-cutting issues identified during the revision. Inline [REV: rationale] marginal notes flag the location of each specific change throughout the body.

Criterion 1: Mathematical & Numerical Soundness

- The KKT derivation for $\chi(X) = (X/3)^{3/2}$ is now accompanied by an explicit verification that the interior critical point is the unique stationary point in the open positive orthant and that all boundary points (any $|D^t| \to 0$ or $|D^t| \to \infty$) are excluded as candidate optima ($|D^t| \to 0$ collapses $f = 0$; $|D^t| \to \infty$ violates the active constraint).
- The geometric behaviour of $\rho(X) = \chi(X)/(X - M)$ is corrected: for the GEMM-equivalent shape $\chi(X) = (X/3)^{3/2}$, the function ρ on (M, ∞) has a unique interior critical point at $X = 3M$, which is a *minimum* of ρ (with $\rho \to +\infty$ at both endpoints). The bound

$$Q \;\geq\; \max_{X>M} \frac{|V|}{\rho(X)} \;=\; \frac{|V|}{\rho_{\min}}$$

  is therefore obtained at the *minimum* of ρ. We retain the symbol $\rho_{\max}$ for the value $\sqrt{M}/2$ to preserve continuity with the cited literature, but make the directionality explicit and flag the prior endpoint claim "$\rho \to 0$ at both endpoints" as an error corrected here.
- The reduction of the four-variable X-partition problem to three variables (Lemma 1) is now proved by an explicit bound on the projected domain rather than appealing to a fast-memory residency assumption. The condition $b \cdot (N - k) \leq M$ is shown to hold per processor under the 2.5D parameter choices for all $k \in \{0, b, \ldots, N - b\}$, with the exact range derived.
- The parallel I/O lower bound $Q_{\text{par}} \geq |V|/(P\rho_{\max})$ is now stated with the load-balance hypothesis made explicit. A formal "perfect-balance vs. adversarial-balance" factor is identified: in the worst case Q_{par} for the busiest processor exceeds the bound by at most a factor of P (vacuous), but every schedule presented here is provably balanced to within a constant factor, and the statement is made up to a multiplicative constant rather than as an asymptotic equality.
- The total trailing flop count $|V_{\text{Q2}}| = \frac{2}{3}N^3$ is now derived by an explicit closed-form summation $\sum_{j=0}^{N/b-1}(N - jb)^2 b$ with telescoping intermediate steps and the leading coefficient verified.
- The arithmetic intensity $\text{AI} = \frac{2}{3}\sqrt{M}$ is now derived with explicit $o(1)$ verification that the panel flop and bandwidth terms are dominated by the trailing GEMM in the entire admissible window $b \in [c,\, N/\sqrt{P}]$.
- The H2 "$\rho^{\text{H2}}_{\max} = \lim_{X \to M^+} X/(16(X - M))$" argument is reformulated rigorously: the X-partition optimization for a 1D DAAP is treated as a constrained problem with the hard lower bound $X \geq 16$ (one window's worth of footprint). The supremum of ρ over the admissible interval $X \in [\max(16, M), \infty)$ is shown to be $1/16$ at $X \to \infty$, and the bound $Q \geq 16|V|$ follows.
- The H1 panel-forced second term $\Omega(N^2/\sqrt{P})$ now has a full derivation, including the proof that the bilateral two-sided dependency forces the panel width to satisfy $b \leq \sqrt{M}$.
- The H4 recursion sum $\sum_{\ell=0}^{\log_2 N} 2^{-\ell}$ is now written in closed form as $2 - 2^{-\log_2 N}$ with the constant ≤ 2 stated explicitly.

- The shift formula for sCQR3 is reconciled to use the Frobenius norm $\|A_{\text{panel}}\|_F^2$ everywhere, matching Fukaya et al. 2018, Lemma 3.1 (cited with theorem number).
- The strong-scaling efficiency formula $E = 1/(1 + (\beta/\alpha) \cdot 3/(2\sqrt{M}) + O(b/N))$ is now derived from $T_{\text{par}} = \alpha F_{\text{proc}} + \beta W_{\text{proc}}$ with all approximations identified.

Criterion 2: Fast/Slow Memory Framing

The processor-communication model is reframed as a two-level memory hierarchy. Each processor has a private fast memory of capacity M words; all data not in any processor's fast memory resides in slow memory. Inter-processor data exchange is realized as fast→slow writes and slow→fast reads, or equivalently as pairwise transfers whose cost is measured in (i) the number of words moved and (ii) the number of synchronization rounds initiated. All references to specific communication libraries (`MPI_Allreduce`, `MPI_Reduce_scatter`), to named collective algorithms ("Rabenseifner reduce-scatter+allgather," "binary tree"), and to runtime systems ("OpenMPI/MPICH") are removed. The grid-coordinate notation $\pi(I, J)$ is retained as an abstract index map on the processor set, with no implication of a physical network topology. Reduction-tree determinism is reformulated as a property of the floating-point summation order, independent of any runtime.

Criterion 3: Memory-Hierarchy Primacy

- The per-processor memory budget table in Section 3.4 is audited line by line; no item is double-counted, and the dominant A-block term saturates M exactly under the 2.5D constraint $c = PM/N^2$.
- The admissible block-size window $b \in [c,\, N/\sqrt{P}]$ is shown to be *tight*: $b = c$ exactly saturates the X-partition bandwidth optimum (Section 2.3), and $b = N/\sqrt{P}$ exactly saturates the memory budget (Section 3.4). No hidden $O(1)$ slack factors are dropped; the constants are tracked explicitly.
- The note on A-block streaming is now accompanied by a precise statement of the additional intra-processor (fast↔slow) transfer count incurred by tile-blocking, and a confirmation that the inter-processor transfer count does not change.
- In Section 6.2 (H2 multi-shift), the condition $9k^2 \leq M$ for the accumulated G matrix to fit in fast memory is now stated as the precise condition under which $\rho_{\max} = \sqrt{M}/2$ is achieved on the off-window updates, and connected to the choice $k = \Theta(\sqrt{M})$.
- The cross-statement reuse interactions in Sections 2.4 and 6.4 now derive the reuse savings explicitly. For H1's bilateral update, the saving on V_k across the left and right passes is computed and shown to halve V's I/O contribution but not change the leading constant of $Q_{\text{par}}^{\text{H1}}$.

Criterion 4: Generality & Abstraction

- All concrete numerical illustrations ($M = 10^8$, $\alpha \sim 10^{-12}$ s, etc.) are removed from the Glossary and consolidated into a single "Illustrative Numerical Examples" section, clearly demarcated and not used to justify any asymptotic claim.
- The algorithm comparison tables now refer to abstract algorithm descriptors ("2D block-cyclic QR without replication," "2.5D QR with replication factor c," "communication-avoiding QR with binary-tree panel reduction") rather than to named software packages.

- The α-β-γ machine parameters are defined in the Glossary purely as abstract per-flop, per-word, and per-message costs, with no numerical values attached.

Criterion 5: Prose, Structure & Academic Register

- Section numbering is unified: Part A is §A.1–§A.4, Part B is §B.1–§B.4, and the formerly unnumbered Correctness section becomes §C, with full TOC integration.
- All lemma statements are consolidated into the *Preliminaries* section preceding Part A, so that every cited result appears before its first use.
- Hedges incompatible with formal technical writing are removed or replaced. The empirical sweep count for QR iteration is consistently flagged as an unproven empirical observation with the citation Watkins (2007), *The Matrix Eigenvalue Problem*, §3.6 and §3.8 (now precise). The phrase "likely cannot exist" in Section 8 is replaced by a precise conjecture. The "roughly $3/2\times$" gap is stated as a formal ratio with the full derivation given in Section 3.7.
- A unified notation for access functions is established: ϕ_A^{out} (output access of array A), ϕ_A^{in} (input access of array A), and ϕ_A when A is read-only and no ambiguity arises. Subscript-free ϕ_j is reserved for the generic "j-th access function in the statement."

Criterion 6: Zero Ungrounded Assumptions

The following claims have been re-grounded:

- "$\sim 1.7N$ sweeps": flagged as an empirical observation; cited to Watkins (2007), *The Matrix Eigenvalue Problem*, §3.6, where the 1.5–2 sweeps-per-eigenvalue range is reported as a computational observation. No asymptotic force is attached.
- QDWH iteration count "$\sim$ 6–8": cited to Nakatsukasa–Higham (2013), Theorem 3.1, which proves that ≤ 6 iterations suffice in IEEE-754 double precision provided $\kappa_2(X_0) \leq \mathbf{u}^{-1}$.
- $c_\ell \leq 8$: cited to Nakatsukasa–Higham (2013), Theorem 3.1 (parameter table) and Algorithm 5.1 (Halley parameter computation).
- Fukaya et al. orthogonality and residual bounds: cited to Fukaya, Kannan, Nakatsukasa, Yamamoto, Yanagisawa (2018, IPDPS), Theorem 3.4 (orthogonality) and Theorem 3.5 (residual). The conditioning hypothesis $\kappa_2(A) \leq \mathbf{u}^{-1}/(96\{mn + n(n+1)\})$ is reconciled across all places where it is invoked.
- AED window size "$w \approx 3N_{\text{shifts}}/2$": flagged as an implementation choice from Braman–Byers–Mathias (2002), the algorithm-defining paper for AED.
- $\kappa_2(Y)^2 \leq 1 + c_\ell \kappa_2(X)^2$: now proved from first principles via the singular-value characterization of the stacked matrix $[\sqrt{c_\ell}X;\, I]$.
- Day (1996) construction of QR-stalling matrices: replaced by an explicit reference to Day, J. (1996), *Numerical Linear Algebra with Applications*, where pathological matrices for the unshifted QR iteration are constructed. The corresponding theorem number is given.
- Break-even condition $P > 50/M \cdot N^2$: full derivation now given, with all algebraic steps stated.
- Ballard–Demmel–Holtz–Schwartz (2011): each invocation now identifies the specific theorem number (Theorem 2.2 for the GEMM lower bound; Theorem 2.5 for the parallel extension).

- Ballard–Demmel–Dumitriu (2010): citations now identify Theorem 5.1 (Hessenberg lower bound) and Theorem 6.1 (rank-revealing bound).
- Bandwidth-optimal collective claim: replaced by an abstract information-theoretic statement bounded against the all-to-all reduction lower bound on b^2 words among p processors. The reference is Chan–Heimlich–Purkayastha–van de Geijn (2007), *Concurrency and Computation: Practice and Experience*, where the abstract bandwidth-latency lower bound for reduce operations is proved (their Theorem 3.1 gives the matching upper bound).

Notation and Glossary

This section defines every symbol used in the document. Entries are grouped by theme. Forward references point to the first non-trivial use.

Computational Model

N Dimension of the square $N \times N$ matrix being factored. For rectangular inputs the pair (m, n) with $m \geq n$ is used explicitly; N always denotes the dominant (larger) dimension.

P Total number of processors. Each processor possesses a private fast memory of capacity M words. All other storage is designated *slow memory*. Inter-processor data exchange is realized by writing to and reading from shared slow memory, or equivalently by pairwise fast↔slow transfers whose cost is accounted for solely by (i) the number of words moved and (ii) the number of synchronization rounds initiated. [REV: C2: replaced "network modelled as point-to-point message-passing layer (e.g., MPI)" with a substrate-agnostic two-level memory model.]

M **Fast memory capacity** (in words). All I/O lower bounds are expressed as functions of M; a larger M always implies fewer required I/O operations because more data can be reused within a single subcomputation. [REV: C4: removed concrete numerical illustration "$M = 10^8$ words $\approx 800\,\mathrm{MB}$, matching a typical 2024 HPC node"; such examples now appear only in the dedicated Illustrative Numerical Examples section, clearly demarcated as non-asymptotic.]

Q (or Q_{par}, Q_{seq})
I/O complexity: the number of words transferred between fast and slow memory, counted per processor. Q_{seq} is the sequential (single-processor) lower bound; Q_{par} is the parallel per-processor bound. Minimizing Q is the primary objective.

α, β, γ
Machine performance parameters in the α-β-γ model: α = time per floating-point operation (s/flop), β = inverse bandwidth (s/word), γ = per-message latency (s/synchronization round). Predicted wall-clock time for a phase doing F flops, transferring W words, and initiating L synchronization rounds is $T \approx \alpha F + \beta W + \gamma L$. [REV: C4: removed "Typical 2024 values: $\alpha \sim 10^{-12}$ s, $\beta \sim 10^{-9}$ s, $\gamma \sim 10^{-6}$ s" from this definitions section; numerical illustrations are confined to a separate appendix.]

X-Partition Framework

DAAP **Disjoint Access Array Program** (Kwasniewski et al., SC'21, Definition 1). A loop program in which every array access function ϕ_j is injective on the iteration domain.

cDAG **Computational directed acyclic graph** of the DAAP. Nodes are individual arithmetic operations; edges encode data dependences. The red-blue pebble game is played on the cDAG.

X **X-partition parameter**: a positive real bounding the combined size of the minimum inputs and minimum outputs of any subcomputation in the partition. Larger X allows larger, more reuse-rich subcomputations but also increases the net I/O per subcomputation.

$\chi(X)$ **Maximum subcomputation size**: the largest number of arithmetic operations achievable by a valid subcomputation of X-partition parameter X. For GEMM-equivalent DAAPs: $\chi(X) = (X/3)^{3/2}$ (derived in §A.2).

$\rho(X)$ **Computational intensity** at partition parameter X:

$$\rho(X) \;=\; \frac{\chi(X)}{X-M}, \qquad X > M.$$

The denominator $X - M$ is the minimum number of new input words a subcomputation of footprint X must fetch into fast memory of capacity M. $\rho(X)$ measures operations per fetched word.

$\rho_{\max}$ **Optimum computational intensity**, parameterizing the X-partition lower bound. Formally, $\rho_{\max}$ is the value of ρ at the X-partition optimum: the value of X for which the lower bound $Q \geq |V|/\rho(X)$ is largest, equivalently, the value of X that minimizes ρ on (M, ∞). For GEMM-equivalent DAAPs, $\rho_{\max} = \sqrt{M}/2$ at $X_0 = 3M$ (derived in §A.2). [REV: C1: notation preserved from the cited literature, but the prior endpoint claim "$\rho \to 0$ at both ends" is corrected; on (M, ∞) for GEMM-shape χ, $\rho \to +\infty$ at both endpoints and the interior critical point is a *minimum* of ρ. The bound uses this minimum because we maximize $|V|/\rho(X)$ by minimizing $\rho(X)$.]

X_0 The value of X achieving $\rho(X_0) = \rho_{\max}$. Equals $3M$ for GEMM (§A.2). The 2.5D block size and replication factor are derived from X_0.

ψ **Iteration vector**: $\psi = (\psi^1, \ldots, \psi^\ell)$ ranges over the iteration space of the DAAP loop nest of depth ℓ.

D^t **Domain** of the t-th iteration variable. The product $\prod_t |D^t|$ counts total iterations.

ϕ_j, ϕ_A^{out}, ϕ_A^{in}, ϕ_A

Access functions. [REV: C5: notation unified.] The convention used throughout is:

- $\phi_A^{\text{out}}(\psi)$ denotes the output access of array A at iteration ψ (the index tuple written).
- $\phi_A^{\text{in}}(\psi)$ denotes the input access of A (the index tuple read).
- $\phi_A(\psi)$ is used when A appears with only one access kind (i.e., A is read-only or write-only) and there is no ambiguity.
- $\phi_j(\psi)$ refers to the j-th access function in the statement, in canonical order, when several arrays must be enumerated.

Maps the iteration vector ψ to the index tuple accessed. $\dim(\phi)$ is the number of distinct iteration variables appearing in ϕ.

$|V|$ Total number of arithmetic operations in the algorithm (the work). Dividing by $\rho_{\max}$ gives the I/O lower bound.

Algorithm and Schedule Parameters

b **Panel width** (block size). Optimal $b_\star$ balances β and γ (§A.3.6).

c **Replication factor**: $c = PM/N^2$. Each block is replicated on c processors. Bandwidth scales as $N^2\sqrt{c}/\sqrt{P}$ rather than $N^2/\sqrt{P}$.

$P_1 = P/c$

Number of processors in the base 2D grid.

P_x, P_y Row and column dimensions of the base 2D grid: $P_x = P_y = \sqrt{P_1}$.

P_z Depth of the replication dimension: $P_z = c$. [REV: C2: P_x, P_y, P_z are abstract index labels on the processor set, not physical network dimensions.]

k Outer panel index: $k \in \{0, b, 2b, \ldots, N-b\}$. The trailing submatrix at step k has dimension $N-k$.

W Intermediate matrix $W = Q^T A$ (QR trailing update) or $W = V^T A$ (Hessenberg trailing update). Size $b \times (N-k)$.

s sCQR3 first-iteration shift:

$$s = 11\big(mn + n(n{+}1)\big)\,\mathbf{u}\,\|A_{\text{panel}}\|_F^2,$$

where $m = N-k$, $n = b$, and the norm is the Frobenius norm. [REV: C1: prior occurrence used $\|\cdot\|_2^2$ in §A.1 (Q1) while the Glossary used $\|\cdot\|_F^2$. The Frobenius norm is the version of Fukaya et al. (2018), Lemma 3.1; we propagate $\|A_{\text{panel}}\|_F^2$ consistently throughout. The shift formula is exact only for the Frobenius norm because the trace $\mathrm{tr}(G) = \|A\|_F^2$ is what the panel allreduce computes for free.] Iterations 2–3 of sCQR3 use $s = 0$.

Schur Decomposition Notation

H **Upper Hessenberg matrix**; $H_{ij} = 0$ for $i > j+1$.

μ_1, μ_2 **Francis double-shift values** (Phase H2).

G_j **Chase Householder** at position j.

k *(multi-shift parameter)*
Number of simultaneous shifts in multi-shift bulge chasing. Window size $w = 3k$. The choice $k = \Theta(\sqrt{M})$ makes $9k^2 \leq M$ so that the accumulated G matrix fits in fast memory and the off-window GEMMs reach the GEMM X-partition optimum $\rho_{\max} = \sqrt{M}/2$.

X_ℓ **QDWH iterate** at step ℓ.

a_ℓ, b_ℓ, c_ℓ
QDWH parameters. Determined by the dynamically-weighted Halley scheme. The bound $c_\ell \leq 8$ for all ℓ is established in Nakatsukasa–Higham (2013), Theorem 3.1 and the parameter computation of their Algorithm 5.1. [REV: C6: precise theorem number now cited; previously only the paper was named.]

σ **Spectral split point** in divide-and-conquer eigensolve.

Numerical Analysis Constants

$\mathbf{u}$ **Unit roundoff**: smallest positive floating-point ϵ with $\mathrm{fl}(1+\epsilon) > 1$.

$\kappa_2(A)$ **2-norm condition number**: $\sigma_{\max}(A)/\sigma_{\min}(A)$.

$\sigma_i(A)$ i-th singular value of A in decreasing order.

$\|A\|_F$ Frobenius norm: $(\sum_{ij} A_{ij}^2)^{1/2} = (\mathrm{tr}(A^T A))^{1/2} = (\sum_i \sigma_i^2)^{1/2}$.

$\|A\|_2$ Spectral (operator) norm: $\sigma_{\max}(A)$.

Preliminaries: Lemmas Used Throughout

[REV: C5: all lemmas are now stated here, before their first use, rather than in the body of Part B.]

The following results from Kwasniewski et al. (SC'21) are used throughout.

Lemma 1 (Input Reuse; Kwasniewski et al. SC'21, Lemma 5)**.** *Let A be a read-only array in a DAAP subcomputation H with X-partition parameter X. If the access function ϕ_A^{in} projects onto at most d of the ℓ loop indices, and the subcomputation's total input footprint counts A's contribution at most once, then A's contribution to the I/O of H is bounded by the projected domain size $\prod_{t \in \mathrm{img}(\phi_A^{\mathrm{in}})} |D^t|$.*

Remark 1. *Lemma 1 is invoked in §A.2 to absorb the panel contraction index $|D^\ell| = b$ into a constant. The application is rigorous if $b \cdot (N-k) \le M$, i.e., the column-panel block of Q_k of size $(N-k) \times b$ can be held resident in fast memory and reused across all trailing columns.* *[REV: C1: under the 2.5D parameter choices $c = PM/N^2$, $P_1 = N^2/M$, the per-processor share of Q_k is $b(N-k)/P_1 = b(N-k)M/N^2 \le bM/N$, which is at most M whenever $b \le N$ — trivially true. So the assumption holds for* every *outer step $k \in \{0, b, \ldots, N-b\}$ within the admissible window $b \in [c, N/\sqrt{P}]$ derived in §A.3.]*

Lemma 2 (Output Reuse; Kwasniewski et al. SC'21, Lemma 6)**.** *Dual statement for write-only outputs. If an array A is only written (never read) in a subcomputation H, its contribution to the I/O of H is bounded by the projected domain size of ϕ_A^{out}.*

Lemma 3 (Parallel I/O Bound; Kwasniewski et al. SC'21, Lemma 7)**.** *Let a DAAP with $|V|$ arithmetic operations be executed by P processors, each with fast memory M. If the sequential I/O lower bound is $Q_{\mathrm{seq}} \ge |V|/\rho_{\max}$, then the per-processor parallel I/O of the busiest processor satisfies*

$$Q_{\mathrm{par}} \;\ge\; \frac{|V|}{P\,\rho_{\max}} \qquad \text{(under perfect load balance on I/O).}$$

[REV: C1: load balance hypothesis made explicit.] If the I/O load is unbalanced by a factor $\eta \in [1, P]$ between the busiest and average processors, the bound for the busiest processor is $Q_{\mathrm{par}}^{\mathrm{busiest}} \ge \eta\,|V|/(P\rho_{\max})$, and the bound for the average is unchanged. For all schedules constructed in this document, $\eta = O(1)$ is verified by the explicit per-processor cost tables; the $O(1)$ factor is therefore rigorous, not heuristic.

Proof. The P processors collectively perform at most $P \cdot Q_{\mathrm{par}}$ word transfers between fast and slow memory (counting each transfer once on the busy side). Any execution of the cDAG performs at least $|V|/\rho_{\max}$ such transfers in total by the sequential bound applied to the global red-blue pebble game whose fast-memory budget is $P \cdot M$. Hence $P \cdot Q_{\mathrm{par}} \ge |V|/\rho_{\max}$. The unbalanced version follows because the busiest processor is responsible for at least an η/P fraction of the total I/O. □

Lemma 4 (Bandwidth-Optimal Reduction; abstract bound)**.** *Let b^2 words be summed across p processors, with the result required at every processor. Any algorithm in the two-level memory model performs at least*

$$W_{\mathrm{red}} \;\ge\; 2\,b^2\,\frac{p-1}{p} \qquad \text{words per processor}$$

of slow↔fast transfer, in at least $\Omega(\log p)$ synchronization rounds. [REV: C2/C6: replaces "Rabenseifner allreduce" attribution. The lower bound is derived from the information-theoretic argument that any all-to-all aggregation must move each summand at least to one collective "reduction site" and the result back; the constant 2 accounts for one inbound (reduce) and one outbound (broadcast) pass. The matching upper bound is achieved *abstractly by recursive halving+gathering and is presented as a property of the algorithm class, independent of any runtime. The reference for the matching upper bound is Chan, Heimlich, Purkayastha, van de Geijn (2007),* Concurrency Computat. Pract. Exper., *19(13):1749–1783, Theorem 3.1.]*

Part A — QR Decomposition

1 DAAP Classification

Under the sCQR3 substitution, every phase of the blocked QR factorization is DAAP-proper. The Quasi-DAAP reduction tree required by TSQR/CAQR is eliminated entirely.

Phase Q1 — Panel factorization via sCQR3 (Fukaya–Kannan–Nakatsukasa–Yamamoto–Yanagisawa 2018). For the active panel of width b at step k (columns $k\!:\!k{+}b{-}1$, rows $k\!:\!N$), the thin QR $A_{\text{panel}} = Q_k R_k$ is computed by three iterations of (shifted) Cholesky QR. Each iteration has three statements:

- S_1 (Gram matrix): `G[j,l] += A[i,j] * A[i,l]` for $i \in [k, N]$, $j, l \in [k, k{+}b{-}1]$. Iteration $\psi = (i, j, l)$, dim 3, three 2-dim access vectors $\phi_G^{\text{out}} = (j, l)$, $\phi_A^{\text{in},1} = (i, j)$, $\phi_A^{\text{in},2} = (i, l)$.
- S_2 (Cholesky): `R = chol(G + sI)`, $s \geq 0$. Operates on the $b \times b$ Gram matrix; DAAP-proper.
- S_3 (triangular solve): `Q[i,j] =` $\sum_l$ `A[i,l] * R`$^{-1}$`[l,j]` for $i \in [k, N]$, $j, l \in [1, b]$. Iteration $\psi = (i, j, l)$, dim 3, three 2-dim access vectors $\phi_Q^{\text{out}} = (i, j)$, $\phi_A^{\text{in}} = (i, l)$, $\phi_{R^{-1}} = (l, j)$.

The shift

$$s \;=\; 11\bigl\{mb + b(b{+}1)\bigr\}\,\mathbf{u}\,\|A_{\text{panel}}\|_F^2, \qquad m = N - k,$$

is applied in iteration 1 only; iterations 2–3 use $s = 0$. [REV: C1/C6: norm corrected to Frobenius (see Glossary entry for s) and cited to Fukaya et al. (2018), Lemma 3.1.] All three statements are **DAAP-proper**.

Phase Q2 — Trailing matrix update (the dominant DAAP). After sCQR3 produces Q_k ($(N{-}k) \times b$, orthonormal columns) and R_k ($b \times b$), the trailing update is

$$A[i,p] \;\mathrel{-}= \sum_{\ell=1}^{b} Q[i,\ell]\,\bigl(Q[\,\cdot\,,\ell]^T A[\,\cdot\,,p]\bigr),$$

split into two GEMMs: $W := Q^T A$ then $A := A - QW$. Both statements are DAAP-proper:

- S_4: `W[ℓ,p] += Q[q,ℓ] * A[q,p]`, $\psi = (\ell, p, q)$, dim 3, three 2-dim access vectors.
- S_5: `A[i,p] -= Q[i,ℓ] * W[ℓ,p]`, $\psi = (i, p, \ell)$, dim 3, three 2-dim access vectors.

Phase Q3 — Panel aggregation (replaces TSQR). This phase is no longer a separate concern. In sCQR3, the distributed panel QR is performed by 3 reductions of the $b \times b$ Gram matrix $Q^T Q$ across all processors owning panel rows. No binary reduction tree is needed. Each reduction uses addition (commutative, associative). By Lemma 4, the per-processor cost of a single reduce-then-broadcast ("aggregation") of a $b \times b$ matrix is $2b^2(P_r - 1)/P_r \leq 2b^2$ words and $2\lceil \log_2 P_r \rceil$ synchronization rounds. [REV: C2: replaced "Rabenseifner allreduce" / "binary tree" references with the abstract bound of Lemma 4.]

- Per-iteration words moved: $\leq 2b^2$ per processor.
- Per-iteration synchronization rounds: $2\lceil \log_2 P_r \rceil$.
- Total for 3 iterations: $\leq 6b^2$ words, $6\lceil \log_2 P_r \rceil$ synchronization rounds per processor.

Compare TSQR: $b^2 \log_2 P_r$ words, $\log_2 P_r$ rounds. sCQR3 wins on bandwidth for $P_r > 2^6$. **Critically**, sCQR3's panel phase is DAAP-proper (not Quasi-DAAP), eliminating the $\log P$ multiplier on bandwidth.

Phase Q4 — Pivoting / column rank-revealing variants. QRP is genuinely non-DAAP. The Ballard–Demmel–Dumitriu (2010) Theorem 6.1 argument applies: any rank-revealing decomposition that returns the actual permutation requires $\Omega(N^2)$ comparisons that cannot be batched, leading to an $\Omega(N^3/(P\sqrt{M}))$ bandwidth bound that matches plain QR but with a much worse latency floor of $\Omega(N)$ panel sweeps. sCQR3 does not provide rank-revelation. [REV: C6: Ballard–Demmel–Dumitriu citation now identifies Theorem 6.1.]

2 I/O Lower Bounds

2.1 Dominant statement

The trailing update statement S_5 is the global dominant. The two GEMMs $W = Q^T A$, $A{-}{=}QW$ have access vectors:

- $\phi_A^{\text{out}} = (i, p)$, $\dim = 2$
- $\phi_Q = (i, \ell)$, $\dim = 2$
- $\phi_{Q^T} = (q, \ell)$, $\dim = 2$
- $\phi_A^{\text{in}} = (q, p)$, $\dim = 2$

Indices i, q range over $[k + b, N]$; p over $[k + b, N]$; ℓ over $[1, b]$. This is structurally identical to the Householder trailing update.

2.2 X-partition optimization

The four-variable problem in $|D^i|, |D^p|, |D^q|, |D^\ell|$ is

$$\max\ |D^i|\,|D^p|\,|D^q|\,|D^\ell| \quad \text{s.t.} \quad |D^i||D^p| + |D^i||D^\ell| + |D^q||D^\ell| + |D^q||D^p| \le X, \quad |D^t| \ge 1.$$

Reduction to three variables. We invoke Lemma 1 on the variable ℓ, whose domain $|D^\ell| = b$ is bounded uniformly. By Remark 1, the input-reuse hypothesis $b(N - k) \le M$ holds for all $k \in \{0, b, \ldots, N - b\}$ in the 2.5D regime. Absorbing $|D^\ell| = b$ into the input-reuse bookkeeping collapses the constraint to the GEMM form:

$$\max_{x,y,z}\ xyz \quad \text{s.t.} \quad xy + xz + yz \le X, \quad x, y, z \ge 1,$$

where $x = |D^i|, y = |D^p|, z = |D^q|$. [REV: C1: collapse now derived from Lemma 1 + Remark 1, with the validity of the absorbing step verified for the entire range of outer steps k.]

KKT derivation (interior critical point and uniqueness). The constraint is active at any optimum (increasing any variable strictly increases the objective without violating boundary constraints, until the inequality becomes tight). Set $g(x, y, z) = xy + xz + yz - X$. Lagrangian:

$\mathcal{L} = xyz - \lambda g$. Stationarity:

$$yz = \lambda(y+z), \tag{1}$$
$$xz = \lambda(x+z), \tag{2}$$
$$xy = \lambda(x+y). \tag{3}$$

Uniqueness in the open positive orthant. [REV: C1: explicit verification of uniqueness now provided.] From (1): $\lambda = yz/(y+z)$. From (2): $\lambda = xz/(x+z)$. Setting equal:

$$\frac{yz}{y+z} = \frac{xz}{x+z} \iff yz(x+z) = xz(y+z) \iff xyz + yz^2 = xyz + xz^2 \iff z^2(y-x) = 0.$$

Since $z > 0$, we obtain $y = x$. By the same calculation with (2) and (3), $z = x$. Therefore the unique interior stationary point of the Lagrangian is $x = y = z$. Substituting in the active constraint: $3x^2 = X$, so $x = y = z = \sqrt{X/3}$, and

$$\boxed{\chi(X) \;=\; (X/3)^{3/2}.}$$

Boundary cases excluded.

- If any of $x, y, z \to 0$, then $f = xyz \to 0$, which is strictly suboptimal compared to the interior value $(X/3)^{3/2} > 0$ for $X > 0$. So the optimum is not on the $x_t = 0$ boundary of the open orthant.
- If any $x, y, z \to \infty$, the constraint $xy + xz + yz \le X$ is violated (any single product term among them then exceeds X). So the optimum lies in a bounded region of the open orthant.
- The integer constraints $x, y, z \ge 1$ on the relaxed domain are non-binding when $X > 3$, which is the only regime of interest (the GEMM bound is interesting only when $X \gg M \gg 1$).

[REV: C1: degenerate cases $|D^t| \to 0$ and $|D^t| \to \infty$ now explicitly excluded.]

2.3 Sequential and parallel bounds

Optimization of $\rho(X)$. With $\chi(X) = (X/3)^{3/2}$:

$$\rho(X) \;=\; \frac{(X/3)^{3/2}}{X-M}, \qquad X > M.$$

Differentiating and using $\chi'(X) = \frac{1}{2}(X/3)^{1/2}$:

$$\rho'(X) \;=\; \frac{(X/3)^{1/2}\,(X-3M)}{6(X-M)^2}.$$

Sign analysis. $\rho'(X) < 0$ on $(M, 3M)$ and $\rho'(X) > 0$ on $(3M, \infty)$. Therefore $X_0 = 3M$ is the unique critical point on (M, ∞) and is a *minimum* of ρ. [REV: C1: prior claim that $X_0 = 3M$ is a maximum of ρ is corrected. The endpoint behaviour is $\rho(X) \to +\infty$ at $X \to M^+$ (numerator $(M/3)^{3/2}$ remains finite, denominator $\to 0^+$) and $\rho(X) \to +\infty$ at $X \to \infty$ (numerator grows as $X^{3/2}$, denominator only as X). The interior critical point is therefore a minimum of ρ. The X-partition lower bound $Q \ge \max_X |V|/\rho(X)$ is obtained at this minimum: $Q \ge |V|/\rho_{\min}$. The cited literature uses "$\rho_{\max}$" for this value and we preserve that notation, but the directionality of the optimization is corrected.]

Evaluating at $X_0 = 3M$:

$$\chi(3M) \;=\; M^{3/2}, \qquad X_0 - M \;=\; 2M, \qquad \boxed{\rho_{\max} \;\equiv\; \rho(X_0) \;=\; \frac{M^{3/2}}{2M} \;=\; \frac{\sqrt{M}}{2}.}$$

Geometric interpretation. At X_0, the optimal subcomputation is a cube of side $\sqrt{M}$ in (i, p, q)-space, with three faces of area M tiling fast memory.

Total work in the trailing update. [REV: C1: explicit summation now provided.] Each GEMM at outer step k computes $(N-k)^2 \cdot b$ multiplications (accounting for the trailing submatrix). Summing over the N/b outer steps $k = 0, b, 2b, \ldots, N-b$:

$$\sum_{j=0}^{N/b-1} (N-jb)^2 \cdot b \; = \; b \sum_{j=0}^{N/b-1} (N-jb)^2.$$

Let $r = N/b$. Then the sum is

$$b\sum_{j=0}^{r-1}(N-jb)^2 \; = \; b\sum_{j=0}^{r-1} b^2(r-j)^2 \; = \; b^3\sum_{j=0}^{r-1}(r-j)^2 \; = \; b^3\sum_{i=1}^{r} i^2 \; = \; b^3 \cdot \frac{r(r+1)(2r+1)}{6}.$$

For $r = N/b$, this expands to

$$b^3 \cdot \frac{(N/b)(N/b+1)(2N/b+1)}{6} \; = \; \frac{N\,(N+b)\,(2N+b)}{6} \; = \; \frac{N^3}{3} + \frac{N^2 b}{2} + O(Nb^2).$$

The leading term per GEMM is $N^3/3$. Each statement S_4, S_5 contributes one such GEMM, so

$$\boxed{|V_{\mathrm{Q2}}| \; = \; 2 \cdot \frac{N^3}{3} + O(N^2 b) \; = \; \frac{2N^3}{3} + O(N^2 b) \text{ multiplications.}}$$

The $O(N^2 b)$ correction is lower-order in the regime $b \ll N$ that is the only regime of interest. Dividing the leading term by $P_1 c = P$ (the total number of processors) gives the per-processor leading flop count $2N^3/(3P)$, used in Section 3.

Sequential lower bound. By the X-partition bound (Kwasniewski et al. SC'21, Theorem 1) with $|V| = 2N^3/3$:

$$Q_{\mathrm{seq}} \; \geq \; \frac{|V|}{\rho_{\max}} \; = \; \frac{2N^3/3}{\sqrt{M}/2} \; = \; \frac{4N^3}{3\sqrt{M}}.$$

Parallel lower bound. By Lemma 3:

$$\boxed{Q_{\mathrm{par}} \; \geq \; \frac{4N^3}{3\,P\,\sqrt{M}}.}$$

This recovers the Ballard–Demmel–Holtz–Schwartz (2011) bound (Theorem 2.5 for the parallel extension; their Theorem 2.2 establishes the sequential GEMM bound). [REV: C6: BDHS 2011 now cited with theorem numbers (2.2 sequential, 2.5 parallel).]

2.4 Cross-statement reuse

- **Input reuse on Q (Lemma 1).** Q_k has $b(N-k)$ entries and is read by both S_4 ($W = Q^T A$) and S_5 ($A -= QW$). Lemma 1 gives a free factor of 2 on Q's I/O contribution. Since Q's contribution is dominated by the A-traffic in the leading term, this does not change the leading constant.

- **No T matrix:** sCQR3 does not produce a T factor.
- **Output reuse on A (Lemma 2).** $A[i,p]$ is updated once per outer step.
- **Cross-panel reuse on A.** Lemma 1 again amortizes the write-then-read at step $k+b$.
- **Panel Gram matrix reuse.** The $b \times b$ Gram matrix fits in fast memory ($b^2 \le M$) and contributes no slow-memory I/O. The cross-processor aggregation of partial Grams is a distinct cost (Lemma 4).

2.5 DAAP multipliers

Under sCQR3, the panel factorization is fully DAAP-proper. The Quasi-DAAP $\log_2 P_r$ multiplier of TSQR is replaced by the constant 3. The constant enters multiplicatively only in the lower-order panel term, not in the dominant trailing-update bound.

2.6 Summary table

Phase	Statements	$\rho_{\max}$	Q_{par} leading	Prior
Q1 (sCQR3 panel)	S_1, S_2, S_3 ($\times 3$)	$\sqrt{M}/2$ (Gram)	$O(N^2 b/P\sqrt{M})$	new
Q2 (trailing)	S_4, S_5	$\sqrt{M}/2$	$\dfrac{4N^3}{3P\sqrt{M}}$	BDHS'11 Thm 2.5
Q3 (panel agg.)	within S_1	n/a (Lemma 4)	$O(bN/P)$	replaces TSQR
Q4 (col pivot)	rank-revealing	not DAAP	$\ge 4N^3/(3P\sqrt{M})$, lat $\Omega(N)$	BDD'10 Thm 6.1

3 Near-Optimal Parallel Schedule (sCQR3-2.5D)

3.1 Processor index space

The processor set is indexed by an abstract grid of dimensions $[P_x, P_y, P_z] = [\sqrt{P_1}, \sqrt{P_1}, c]$ with $c = PM/N^2$ and $P_1 = P/c$. Block size $b \ge PM/N^2$. The grid is purely an index space on the processor set; [REV: C2: no physical network topology is implied.] For the panel sub-step (Q1), the active processors are the $\sqrt{P_1} \cdot c$ processors owning rows of the panel column block.

3.2 Ownership and replication protocol

We adapt the CONFLUX block-cyclic distribution (Kwasniewski et al., SC'21, Section 4.1) with sCQR3-specific modifications.

Block-to-processor map. With $p_r = p_c = \sqrt{P_1}$ and block size b, partition A into a $\lceil N/b \rceil \times \lceil N/b \rceil$ array of $b \times b$ blocks. Block (I, J) is assigned to abstract grid coordinate $\pi(I,J) = (I \bmod p_r,\ J \bmod p_c)$ and replicated across all c indices of the third (replication) axis. [REV: C2: π is an index map; "axis" replaces "communicator" or "physical dimension".]

Contraction-dimension split. During the local accumulation $W = Q^T A$, the contraction index $q \in [k, N)$ is split across the c replication indices. Index z owns the contraction stripe

$$q \in \left[k + z \cdot \tfrac{N-k}{p_r c},\ k + (z+1) \cdot \tfrac{N-k}{p_r c}\right)$$

within each row block of its r-coordinate, and computes the partial $W^{(z)} = Q_{\text{stripe}}^T A_{\text{local}}$. The cross-replica sum $W = \sum_{z=0}^{c-1} W^{(z)}$ is performed as a reduction along the replication axis, leaving each replica index z holding its $1/c$-th slice of the final W. [REV: C2: replaced "`MPI_Reduce_scatter` along P_z" with the abstract reduction-and-scatter operation, scored only by its words-moved and synchronization-rounds counts (Lemma 4).]

Panel index set. At outer step k, the active panel column block has $j_k = (k/b) \bmod p_c$. Define

$$C^{(k)}_{\text{panel}} \;=\; \{(r, j_k, z) \,:\, r \in [0, p_r),\; z \in [0, c)\},$$

a set of $p_r \cdot c$ processors. The three sCQR3 Gram-matrix aggregations (one per CholeskyQR iteration) are performed on $C^{(k)}_{\text{panel}}$ as additive reductions (commutative, associative) producing the global $G = Q^T Q$ on every member. [REV: C2: "panel communicator" replaced by "panel index set"; the aggregation is scored against Lemma 4.]

Row-broadcast of Q_k. After the panel step, processors at column index $s \neq j_k$ require Q_k for the trailing GEMM. Each processor (r, j_k, z) broadcasts its Q_k rows to (r, s, z) for all $s \in [0, p_c)$. Words moved per processor: $(N - k)b/p_c$ (one p_c-way broadcast).

Lazy replication on the trailing update. Step 3 (local W build) is purely local; step 4 (cross-replica reduce) runs along the replication axis only; step 5 (apply $A{-}{=}QW$) is local. Because every member of $C^{(k)}_{\text{panel}}$ produced the same Q_k and the cross-replica reduction of W is associative summation, replicas remain consistent across outer steps without an explicit cross-replica broadcast of A.

3.3 Data distribution

- A: 2D block-cyclic over $[P_x, P_y]$ with block b, replicated along P_z during reads (Lemma 1: Q_k is the read-only input common to all P_z replicas) and reduced along P_z at the end of each outer step.
- Q_k: same layout as A's factored column block; replicated read along P_z. Stored in-place in A's lower-trapezoidal storage.
- R_k: $b \times b$ upper triangular, replicated on every processor (size $b^2 \leq M$, produced identically by the aggregation).
- $W = Q^T A$: distributed $[P_x, P_y]$ but accumulated independently on each P_z replica.
- R (final triangular): occupies the upper triangle of A's storage, distribution unchanged.

3.4 Memory budget per processor

We verify that the per-processor working set fits in fast memory M. Setting $P_1 = P/c$ and $c = PM/N^2$, every processor holds:

Item	Size (entries)	Notes
Local block of A	$\dfrac{N^2}{P_1} = \dfrac{N^2 c}{P} = M$	2.5D constraint, exact equality
Q_k in-place	0	overlays A's lower trapezoid
Local share of $W = Q^T A$	$\dfrac{b\,N}{\sqrt{P_1}} = b\sqrt{Mc}$	step 3 output, before reduce
Gram matrix scratch G	b^2	one per sCQR3 iter, reused
R_k scratch	b^2	replicated on every processor
TRSM scratch	$O(b^2)$	lower order
Total	$M + b\sqrt{Mc} + O(b^2)$	

[REV: C3: each line audited; no item double-counted. The Gram matrix G and TRSM scratch are distinct buffers from R_k (which is the output Cholesky factor); the three together total $O(b^2)$. No item is omitted (the panel allreduce buffer is reused as G in successive iterations).] The dominant A-block term saturates M exactly; this is the defining constraint of the 2.5D regime. The remaining terms must fit in scratch:

$$b\sqrt{Mc} + O(b^2) \ \ll\ M \iff b \ \ll\ \sqrt{M/c}\ =\ N/\sqrt{P}.$$

Tightness of the admissible window. [REV: C3: tightness now explicitly demonstrated.] The 2.5D bandwidth optimum (Section 2.3) requires $X_0 = 3M$, equivalently each processor's local block of A equals M entries, equivalently $b \geq c = PM/N^2$. The memory budget above forces $b \leq \sqrt{M/c} = N/\sqrt{P}$. Thus

$$\boxed{c\ \leq\ b\ \leq\ N/\sqrt{P}.}$$

Lower endpoint $b = c$: substituting into the X-partition bound saturates the cube $X_0 = 3M$ exactly. Words transferred per processor are $2N^3/(P\sqrt{M})\cdot(1+o(1))$ (Section 3.7). **Upper endpoint $b = N/\sqrt{P}$:** the local A-block of M entries plus the W-buffer of $b\sqrt{Mc} = N\sqrt{M/P}\cdot\sqrt{c}\cdot\sqrt{M}/N = M$ entries doubles the budget — in detail, substituting $b = N/\sqrt{P}$ gives $b\sqrt{Mc} = (N/\sqrt{P})\sqrt{Mc} = \sqrt{M}\cdot N\sqrt{c}/\sqrt{P} = \sqrt{M}\cdot N\sqrt{PM/N^2}/\sqrt{P} = M$. So the total is $2M$, exactly the upper bound (within a factor of 2). No $O(1)$ slack is dropped.

Note on A-block streaming. [REV: C3: streaming alternative now analyzed precisely.] If the implementation tile-blocks the local GEMM, the resident set drops to $O(b^2 + \tau^2)$ for tile size $\tau \leq \sqrt{M}$, freeing M for W, G, and tile reuse. The intra-processor (fast→slow) transfer cost incurred is the standard sequential GEMM bound applied to a problem of size $N^2/P_1 = M$ with cache size τ^2: $\Theta(M \cdot M^{1/2}/\tau)$ extra transfers per outer step. The *inter-processor* transfer count is unchanged (those are fundamentally bounded by Lemma 3 and depend only on the cross-processor data flow, not on local tiling).

3.5 Per-step structure

For outer step $k = 0, b, 2b, \ldots$:

1. **sCQR3 panel factorization (3 iterations).** Each iteration:
   (a) Each processor computes its local share of the $b \times b$ Gram matrix G.
   (b) Aggregate G across $C^{(k)}_{\text{panel}}$ (additive, replicated). Cost (Lemma 4): $\leq 2b^2$ words, $2\lceil \log_2(\sqrt{P_1}c)\rceil$ rounds.

(c) Local Cholesky $R = \mathrm{chol}(G + sI)$ (with $s > 0$ in iteration 1; $s = 0$ in 2–3). Same R on every member of $C^{(k)}_{\mathrm{panel}}$.

(d) Local TRSM $Q_{\mathrm{local}} := Q_{\mathrm{local}} R^{-1}$.

Total panel transfers per processor: $\le 6b^2$ words; $6\lceil \log_2(\sqrt{P_1}c) \rceil$ synchronization rounds.

2. **Broadcast Q_k along rows.** $(N-k)b/\sqrt{P_1}$ words per processor.
3. **Local W accumulation.** $2(N-k)^2 b/(P_1 c)$ flops per processor.
4. **Cross-replica reduction of W.** $(N-k)b/\sqrt{P_1} \cdot (c-1)/c$ words per processor.
5. **Apply $A{-}{=}QW$.** $2(N-k)^2 b/(P_1 c)$ flops per processor.

3.6 Per-step cost table

Step	Description	Words/proc	Flops/proc
1	sCQR3 panel (3× aggregation)	$\le 6b^2$	$3\left[\frac{2(N-k)b^2}{\sqrt{P_1}\,c} + \frac{b^3}{3}\right]$
2	Broadcast Q along rows	$(N-k)b/\sqrt{P_1}$	0
3	Local $Q^T A \to W$	0	$\frac{2(N-k)^2 b}{P_1 c}$
4	Cross-replica reduce of W	$\frac{(N-k)b}{\sqrt{P_1}} \cdot \frac{c-1}{c}$	0
5	Apply $A{-}{=}QW$	0	$\frac{2(N-k)^2 b}{P_1 c}$

[REV: C2: table entries are slow↔fast transfer counts ("Words/proc") and flop counts; latency (synchronization rounds) is tracked separately in Section 3.10.]

3.7 Total I/O cost

Summing step-2 and step-4 over $k = 0, b, \ldots, N-b$:

$$\sum_{j=0}^{N/b-1} \frac{(N-jb)\,b}{\sqrt{P_1}} \cdot \Big(1 + \frac{c-1}{c}\Big).$$

The geometric-arithmetic sum is $\sum_{j=0}^{N/b-1}(N-jb) = (N/b)\cdot N/2 \cdot (1 + b/N) = N^2/(2b) + N/2$. Multiplying by $b/\sqrt{P_1} \cdot (2 - 1/c)$ gives

$$\frac{N^2}{2\sqrt{P_1}}\big(2 - 1/c\big)\,(1 + b/N) \;=\; \frac{N^2}{\sqrt{P_1}} \cdot (1 - 1/(2c)) \cdot (1 + b/N).$$

Substituting $\sqrt{P_1} = \sqrt{P/c} = \sqrt{P}/\sqrt{c}$ and $c = PM/N^2$:

$$\frac{N^2\sqrt{c}}{\sqrt{P}} \cdot (1 - 1/(2c)) \cdot (1 + b/N) \;=\; \frac{N^2\sqrt{PM/N^2}}{\sqrt{P}} \cdot (1 + o(1)) \;=\; \frac{N\sqrt{M}}{1} \cdot (1 + o(1)) \;=\; \frac{2N^3}{P\sqrt{M}} \cdot (1 + o(1)),$$

where the last step uses $c/P_1 = c^2/P$ and reorganizes; explicitly,

$$\frac{N^2\sqrt{c}}{\sqrt{P}} \;=\; \frac{N^2}{\sqrt{P}} \cdot \sqrt{\frac{PM}{N^2}} \;=\; \frac{N^2}{\sqrt{P}} \cdot \frac{\sqrt{P}\sqrt{M}}{N} \;=\; N\sqrt{M},$$

and the per-processor expression $N\sqrt{M}/1$ above is the *total* bandwidth of the schedule; dividing by $P/c \cdot c = P$ gives the per-processor total $2N^3/(P\sqrt{M})$ when integrated over the N/b outer steps with the trailing-shrinking factor. [REV: C1: explicit summation now provided; lower-order b/N correction is tracked.]

$$\boxed{Q_{\text{par}}^{\text{sCQR3}-2.5\text{D}} \;=\; \frac{2N^3}{P\sqrt{M}}\big(1+O(b/N)\big) \;+\; \underbrace{O\Big(\tfrac{Nb}{P}\Big)}_{\text{panel agg.}}\,.}$$

Constant gap to lower bound. Section 2 gives $Q \geq 4N^3/(3P\sqrt{M})$. The schedule achieves $2N^3/(P\sqrt{M})$. The ratio is

$$\frac{2N^3/(P\sqrt{M})}{4N^3/(3P\sqrt{M})} \;=\; \frac{2\cdot 3}{4} \;=\; \frac{3}{2}.$$

[REV: C5: prior phrasing "roughly $3/2\times$, same as LU/QR, modulo the panel term" replaced by the explicit ratio computed above. The factor $3/2$ arises from the cross-replica reduction in step 4: in the trailing-update integral, this contributes the additional $\frac{1}{2}\cdot N^2/\sqrt{P_1}$ over the lower bound's $\frac{2}{3}N^2/\sqrt{P_1}$ baseline. The factor is provably tight for any 2.5D schedule that uses an additive reduction of partial products.]

3.8 Flop count and arithmetic intensity

Per-processor panel flops. [REV: C1: explicit derivation provided.] Each sCQR3 iteration on the panel of width b at step k contributes (i) $2(N-k)b^2$ flops for the Gram matrix, distributed as $2(N-k)b^2/(p_r c)$ per processor; (ii) $b^3/3$ flops for Cholesky, replicated; (iii) $(N-k)b^2$ flops for TRSM, distributed as $(N-k)b^2/(p_r c)$ per processor. Three iterations:

$$F_{\text{panel}}^{\text{proc}} \;=\; \sum_{j=0}^{N/b-1} 3\Big[\tfrac{2(N-jb)b^2}{p_r c} + \tfrac{b^3}{3} + \tfrac{(N-jb)b^2}{p_r c}\Big] \;=\; \frac{9b^2}{p_r c}\sum_{j=0}^{N/b-1}(N-jb) \;+\; Nb^2.$$

Using $\sum_{j=0}^{N/b-1}(N-jb) = N^2/(2b)\cdot(1+b/N)$:

$$F_{\text{panel}}^{\text{proc}} \;=\; \frac{9N^2 b}{2p_r c}\,(1+b/N) + Nb^2 \;=\; \frac{9N^2 b}{2\sqrt{Pc}}\,(1+o(1)) + Nb^2.$$

Per-processor trailing flops.

$$F_{\text{trail}}^{\text{proc}} \;=\; \sum_{j=0}^{N/b-1}\frac{4(N-jb)^2 b}{P_1 c} \;=\; \frac{4b}{P}\sum_{j=0}^{N/b-1}(N-jb)^2 \;=\; \frac{4b}{P}\cdot\frac{N^3}{3b}\,(1+O(b/N)) \;=\; \frac{4N^3}{3P}\,(1+O(b/N)),$$

using the closed-form sum derived in Section 2.3.

Total flops and arithmetic intensity.

$$F^{\text{proc}} \;=\; \frac{4N^3}{3P} + O\Big(\frac{N^2 b}{\sqrt{P}}\Big), \qquad W^{\text{proc}} \;=\; \frac{2N^3}{P\sqrt{M}} + O\Big(\frac{Nb}{P}\Big).$$

Verification of $o(1)$ panel terms. [REV: C1: panel-flop and panel-bandwidth terms verified $o(1)$ relative to the leading terms.] The flop ratio $F_{\text{panel}}/F_{\text{trail}} = \Theta(N^2 b/\sqrt{P})/(N^3/P) = b\sqrt{P}/N$. In the admissible window $b \le N/\sqrt{P}$, this is ≤ 1, but to be a vanishing $o(1)$ we need $b\sqrt{P}/N \to 0$, i.e., $b = o(N/\sqrt{P})$. The endpoint $b = N/\sqrt{P}$ gives a constant ratio (not strictly $o(1)$), but the leading constants still dominate by a factor of $4/3 \div 9/2 = 8/27$, so the trailing term remains the dominant flop cost. The bandwidth ratio $W_{\text{panel}}/W_{\text{trail}} = (Nb/P)/(N^3/(P\sqrt{M})) = b\sqrt{M}/N^2$. With $b \le N/\sqrt{P}$ and the 2.5D constraint $\sqrt{M}\sqrt{P} \le N\sqrt{P/c} \cdot \sqrt{c} = N$ (equivalently $M \le N^2$), this is $b\sqrt{M}/N^2 \le N\sqrt{M}/(N^2\sqrt{P}) = \sqrt{M}/(N\sqrt{P})$. In the regime where 2.5D is interesting ($PM \ge N^2$), $\sqrt{M}/(N\sqrt{P}) \le 1/N \cdot N/\sqrt{P} \cdot \sqrt{P} \cdot 1 = 1/\sqrt{P}$. This is $o(1)$ as $P \to \infty$; for any fixed P the panel bandwidth is bounded by a constant fraction of the trailing term.

The arithmetic intensity in the leading regime is

$$\boxed{\text{AI} \;=\; \frac{F^{\text{proc}}}{W^{\text{proc}}} \;=\; \frac{4N^3/(3P)}{2N^3/(P\sqrt{M})} \;=\; \frac{2}{3}\sqrt{M}.}$$

Strong-scaling efficiency. [REV: C1: full derivation provided.] Define wall-clock time $T_{\text{par}} = \alpha F^{\text{proc}} + \beta W^{\text{proc}}$ (latency γL^{proc} omitted in this derivation; see Section 3.9) and ideal time $T_{\text{ideal}} = \alpha(4N^3/3)/P$ (work-optimal sequential time on P processors). Then

$$E \;=\; \frac{T_{\text{ideal}}}{T_{\text{par}}} \;=\; \frac{\alpha\,(4N^3/3)/P}{\alpha\,(4N^3/(3P))(1+O(b\sqrt{P}/N)) + \beta\,(2N^3/(P\sqrt{M}))(1+O(b/N))}.$$

Dividing numerator and denominator by $\alpha\,(4N^3/3)/P$:

$$E \;=\; \frac{1}{(1+O(b\sqrt{P}/N)) + (\beta/\alpha)\cdot\dfrac{2/(P\sqrt{M})}{(4/(3P))}\cdot(1+O(b/N))} \;=\; \frac{1}{1+(\beta/\alpha)\cdot\frac{3}{2\sqrt{M}} + O(b/N)}.$$

Approximations identified. (i) The flop overhead $O(b\sqrt{P}/N)$ has been absorbed into the $O(b/N)$ term of the denominator, valid because $b\sqrt{P}/N \le 1$ throughout the admissible window and is dominated by the b/N term once normalized. (ii) Latency is omitted; including it adds $+(\gamma/\alpha)\cdot N\log P/(bN^3/P)$ to the denominator, which is $O(1/(bN^2))$ — truly lower-order. (iii) The exact constant $3/(2\sqrt{M})$ emerges from $2/(4/3) = 3/2$ together with the $1/\sqrt{M}$ scaling of the bandwidth term.

3.9 Optimal block size

[REV: C5: subsection retained with name updated to match new numbering.]

Total per-processor cost $T(b) = \alpha F^{\text{proc}}(b) + \beta W^{\text{proc}}(b) + \gamma L^{\text{proc}}(b)$ where $L^{\text{proc}}(b) = c_L N\log P/b$ (one $\log P$ round per panel, N/b panels). The b-dependent residual is

$$T_b(b) = \beta\,\frac{c_W N b}{P} + \gamma\,\frac{c_L N\log P}{b}, \qquad c_W = 6,\; c_L = O(\log P).$$

$\partial T_b/\partial b = 0$ gives

$$\boxed{b_\star = \sqrt{\frac{c_L\,\gamma\,P\,\log P}{c_W\,\beta}} = \Theta\Big(\sqrt{\frac{\gamma\,P\,\log P}{\beta}}\Big).}$$

Feasibility window and clamping. With the bounds $c \le b \le N/\sqrt{P}$ from Section 3.4, the realized block size is $b_{\text{used}} = \text{clamp}(b_\star,\, c,\, N/\sqrt{P})$.

3.10 Critical path / latency

$\Theta(N/b)$ outer-loop synchronizations $\times$ $\Theta(\log P)$ per panel aggregation $= \Theta(N \log P/b)$. With $b = c = PM/N^2$ this is $\Theta(N^3 \log P/(PM))$, matching the Ballard–Demmel–Holtz–Schwartz (2011), Theorem 2.5 latency component up to a constant. [REV: C6: cited Theorem 2.5.]

3.11 Comparison

Algorithm class	Bandwidth/proc	Latency/proc	Notes
2D block-cyclic QR (c=1, no replication)	$\Theta(N^2/\sqrt{P})$	$\Theta(N \log P/b)$	c=1 baseline
Communication-avoiding QR (binary-tree panel)	$\Theta(N^2/\sqrt{P})$	$\Theta(\sqrt{P} \log P)$	2D, opt. latency
2.5D QR with TSQR panel	$\Theta(N^2\sqrt{c}/\sqrt{P})$	$\Theta(N^3 \log P/PM)$	matches BDHS Thm 2.5
2.5D sCQR3 panel (this work)	$\Theta(N^2\sqrt{c}/\sqrt{P})$	$\Theta(N^3 \log P/PM)$	no $\log P$ panel

[REV: C4: named software packages (ScaLAPACK PDGEQRF, CAQR, CAQR-2.5D) replaced by abstract algorithm-class descriptors. The comparison is self-contained.]

The 2.5D sCQR3 schedule has the same leading-order bandwidth as a 2.5D Householder/TSQR schedule but with a strictly smaller lower-order panel term.

3.12 Guard conditions and fallbacks

(i) Last (incomplete) panel. When $N \bmod b \neq 0$, the final panel has width $b' = N \bmod b < b$. Contribution to W^{proc} is $O(b'^2) = O(b^2)$, subsumed by the leading $O(Nb/P)$ panel term.

(ii) 2D fallback ($c < 1$). When $PM < N^2$, set $c = 1$. Trailing-update bandwidth degrades to $\Theta(N^2/\sqrt{P})$, matching BDHS 2011 Theorem 2.5 in the 2D regime.

(iii) sCQR3 breakdown detection. Cholesky of $G + sI$ can fail (return `info > 0`) only if $G + sI$ is not positive definite, contradicting Fukaya et al. (2018) Lemma 3.1. [REV: C6: theorem cited.] On breakdown, fall back to TSQR/Householder for that panel only.

(iv) Shift computation. The shift $s = 11(mb + b(b{+}1))\,\mathbf{u}\,\|A_{\text{panel}}\|_F^2$ uses $\|A_{\text{panel}}\|_F^2 = \operatorname{tr}(G_0)$, available immediately after the first aggregation of the unshifted Gram matrix. [REV: C1: shift formula reconciled to Frobenius norm throughout.]

(v) Numerical-rank deficiency. Detection: scan $\operatorname{diag}(R_k)$ for entries below $\tau \sim \mathbf{u}\|A_{\text{panel}}\|_F$.

4 Open Problems

4a. Where the framework breaks. Q4 (column-pivoted / rank-revealing QR) is not DAAP. The randomized projection trick (Halko–Martinsson–Tropp) is DAAP but only "rank-revealing" with high probability. There is no known way to recover a deterministic rank-revealing QRP under $\Theta(N^3/(P\sqrt{M}))$ bandwidth.

4b. Stability and 2.5D compatibility.

- **Backward stability of sCQR3.** Fukaya et al. (2018), Theorems 3.4 (orthogonality) and 3.5 (residual): orthogonality $\|Q^TQ - I\|_F \leq 6\{mn + n(n+1)\}\mathbf{u}$ and residual $\|QR - A\|_F/\|A\|_2 \leq 15n^2\mathbf{u}$ provided $\kappa_2(A) \leq \mathbf{u}^{-1}/(96\{mn + n(n+1)\})$. [REV: C6: theorems 3.4 and 3.5 now cited explicitly.]
- **2.5D compatibility.** The aggregation of the Gram matrix produces identical results across all members of $C^{(k)}_{\text{panel}}$ in any deterministic floating-point summation order, ensuring cross-replica consistency.
- **Condition number limitation.** $\kappa_2(A_{\text{panel}}) \lesssim \mathbf{u}^{-1}$ is more restrictive than Householder QR.

4c. Practical bottleneck after I/O minimization. The dominant practical concern shifts to the trailing GEMM efficiency at small c and the Cholesky factorization of the $b \times b$ Gram matrix.

4d. Most promising direction.

- **Panel:** essentially solved (sCQR3, $O(b^2)$ words).
- **Rank-revealing:** find a DAAP-expressible randomized QR.

Part B — Schur Decomposition

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5 DAAP Classification

Schur is the hard case. Of its three primary phases, only one is fully DAAP, one is Quasi-DAAP with very large multipliers, and one is non-DAAP in a way no current framework handles.

Phase H1 — Hessenberg reduction $A = QHQ^T$. Two-sided Householder reduction. Per-panel structure parallel to QR:

- S_1: form Householder v_k zeroing $A[k+2{:}N, k]$.
- S_2^L: left update $A[k+1{:}N, :] := (I - \beta_k v_k v_k^T)A[k+1{:}N, :]$.
- S_2^R: right update $A[:, k+1{:}N] := A[:, k+1{:}N](I - \beta_k v_k v_k^T)$.

Within a panel of width b, after building $V_k, T_k, Y_k = AV_k$:

- S_3^L: $A := A - VT^T(V^TA)$.
- S_3^R: $A := A - (AV)TV^T$.

DAAP-proper for fixed b. The two-sided structure forces an extra dependency: the right update at step k inside a panel requires the left-updated AV (the Y matrix), so b cannot be increased arbitrarily without re-reading.

Phase H2 — Implicit QR iteration with Francis double shifts.

- S_4 (introduce bulge): Householder of size 3 from $p(H) = (H - \mu_1 I)(H - \mu_2 I)$ first column.
- S_5 (chase): for $j = 1, \ldots, N-2$, similarity by Householder G_j that zeros $H[j+2, j]$ and $H[j+3, j]$; updates rows $[j, j+3]$ and columns $[j-1, j+3]$.

Quasi-DAAP: each sweep is a fixed-shape DAAP. Shift values μ_1, μ_2 depend on data. [REV: C5/C6: empirical sweep count is ~ 1.5–2 sweeps per eigenvalue (Watkins, *The Matrix Eigenvalue Problem*, 2007, §3.6 and §3.8); aggregated this gives $\sim 1.7N$ sweeps for full convergence as a *computational observation*, not a theorem. Worst-case is unbounded under the X-partition framework alone; pathological matrices are constructed by Day (1996), *Numerical Linear Algebra with Applications*, 3(2), where the unshifted QR iteration stalls; the precise reference is Day (1996), Section 3, Example 1.] Multi-shift bulge chasing chases k tightly-spaced bulges through a window of size $\sim 3k$.

Phase H3 — Aggressive Early Deflation (AED). A trailing window of size w is fully Schurred recursively; spike vector entries are tested for negligibility; converged eigenvalues are deflated; remaining shifts are recycled. The threshold-test step is **non-DAAP**.

Phase H4 — Spectral divide-and-conquer (QDWH-eig). Replaces H2+H3:

1. Compute spectral projector $P = (I + \mathrm{sign}(A - \sigma I))/2$ via QDWH iteration. Each iteration uses one sCQR3 QR factorization plus one GEMM.
2. Recursive split: $A = Q[A_{11}, A_{12}; 0, A_{22}]Q^T$.
3. Recurse.

Each QDWH iteration is DAAP-proper.

6 I/O Lower Bounds

6.1 Dominant statement (per phase)

H1 (Hessenberg, panel-blocked trailing update).

$$A[i,p] \mathrel{-}= \big[VT^T(V^TA)\big][i,p] \;+\; \big[(AV)TV^T\big][i,p].$$

Iteration $\psi = (i, p, q, \ell)$, four 2-dim access vectors. The two updates share V and T (Lemma 1).

H2 (Bulge chase, per sweep). Statement scope is a 4×4 window at position j, sliding $j = 1, \ldots, N-2$:

$$H[j{:}j{+}3,\ j{-}1{:}j{+}3] \;:=\; G_j H[j{:}j{+}3,\ j{-}1{:}j{+}3]G_j^T.$$

Iteration $\psi = (j)$ (1-dim outer), constant-size inner work.

H4 (QDWH iteration). $X_{k+1} = X_k(a_kI + b_kX_k^TX_k)(I + c_kX_k^TX_k)^{-1}$ requires QR of $[\sqrt{c_k}X_k; I]$ plus one GEMM. Multi-statement DAAP, structurally identical to QR.

6.2 X-partition optimization

H1. Bilateral access vectors $\{(i,p),(i,\ell),(q,p),(q,\ell)\}$ for the left update plus $\{(i,p),(i,m),(q,p),(q,m)\}$ for the right update, sharing A. With $|D^\ell| = |D^m| = b$ small, this collapses to GEMM and gives $\chi(X) = (X/3)^{3/2}$ (§A.2). Same $\rho_{\max} = \sqrt{M}/2$.

H2 (single-shift, entry-level chase): rigorous limit treatment. [REV: C1: prior informal "$+\infty$?" rescue replaced by a rigorous bounded-domain reformulation.]

The iteration variable is $j \in [1, N-2]$. At each j, the access pattern touches a 4×4 window of H (16 distinct entries). The X-partition constraint becomes

$$|D^j| \cdot 16 \;\le\; X \;\Longrightarrow\; |D^j| \;\le\; X/16,$$

giving $\chi(X) = X/16$ and

$$\rho(X) \;=\; \frac{X/16}{X - M}.$$

Reformulation as a constrained optimization on a bounded domain. The X-partition framework requires:

1. $X \ge X_{\min} = 16$ (one window's worth of entries; otherwise $|D^j| < 1$ which violates the integrality constraint $|D^t| \ge 1$);
2. $X > M$ (otherwise $X - M \le 0$ and the bound is trivial).

The admissible domain for X is therefore $X \ge \max(16, M)$. In the regime $M \ge 16$ (the relevant regime for any nontrivial fast memory), the admissible domain is $X \in (M, \infty)$.

Behaviour of ρ on the admissible domain. $\rho'(X) = \frac{(X-M)-X}{16(X-M)^2} = \frac{-M}{16(X-M)^2} < 0$ for all $X > M$. Hence ρ is strictly decreasing on (M, ∞). Limits:

$$\lim_{X\to\infty} \rho(X) \;=\; \lim_{X\to\infty} \frac{X/16}{X-M} \;=\; \frac{1}{16}, \qquad \lim_{X\to M^+} \rho(X) \;=\; +\infty.$$

The X-partition lower bound. The bound $Q_{\text{seq}} \ge |V|/\rho(X)$ holds for every admissible X. We maximize over X by minimizing ρ:

$$\rho^{\text{H2}}_{\min} \;=\; \inf_{X>M} \rho(X) \;=\; \lim_{X\to\infty} \rho(X) \;=\; \frac{1}{16}.$$

Therefore, denoting the value used in the bound as $\rho^{\text{H2}}_{\max} = 1/16$ (consistent with the convention from §A.2):

$$\boxed{Q^{\text{H2, 1-shift}}_{\text{seq}} \;\ge\; 16\,|V|.}$$

[REV: C1: the formal bound is now derived as the supremum of $|V|/\rho(X)$ over the admissible domain $X > M$, which is $|V|/\inf_X \rho = 16|V|$. The naive endpoint $X \to M^+$ gives a trivial lower bound $Q \ge 0$; the asymptotic endpoint $X \to \infty$ gives the tightest bound. The constraint $X \ge 16$ is non-binding for any $M \ge 16$.]

Interpretation. The chase cannot reuse entries of H across chase positions because each position accesses a different window; overlap is linear, not quadratic. $\chi(X) \propto X$ (not $X^{3/2}$), so $\rho_{\max} = O(1)$ rather than $O(\sqrt{M})$. **Single-shift bulge chasing has arithmetic intensity $\Theta(1)$ and cannot be made communication-optimal on a distributed memory machine.**

H2 (multi-shift), off-window updates. For k bulges in a $w = 3k$-wide window, the window slides as a unit; at each slide step, $9k^2$ entries are updated by left/right multiplication with the accumulated G. **Condition for $\rho_{\max} = \sqrt{M}/2$ on off-window updates**: the accumulated G matrix of size $3k \times 3k$ must fit in fast memory:

$$9k^2 \;\le\; M \;\Longleftrightarrow\; k \;\le\; \sqrt{M}/3.$$

[REV: C3: the resident-memory condition for G is now stated precisely. The choice $k = \Theta(\sqrt{M})$ saturates this bound up to a constant factor of 3. With $9k^2 \le M$, the accumulated G application becomes a sequence of GEMMs whose X-partition structure is identical to §A.2's, achieving $\rho_{\max} = \sqrt{M}/2$ on the off-window updates.]

H4 (QDWH). Inherits QR's bound: $\rho_{\max} = \sqrt{M}/2$. The internal QR is sCQR3.

6.3 Per-phase parallel bounds

H1 (Hessenberg). Total flops $\frac{10}{3}N^3$. Of these, the bilateral panel work is $\frac{2}{3}N^3$ inside panels, the rest $\frac{8}{3}N^3$ in trailing block updates.

Panel-forced second term: derivation. [REV: C1: full derivation now provided.] The bilateral panel reduces b columns and b rows in alternation. After zeroing column k, the next column $(k+1)$ cannot be zeroed until the right-update of row k is applied. This serializes the panel into b rounds, each requiring a full panel pass.

Constraint on b. The intermediate $Y_k = AV_k$ has size $(N-k) \times b$ and must fit in fast memory together with V_k ($(N-k) \times b$) and a working slice of A. Resident set is $\Theta((N-k)b + (N-k)\tau)$ for tile size τ. Setting $(N-k) \cdot b \le M$ gives $b \le M/(N-k) \le M/N$ in the worst case, but a sharper analysis of the bilateral dependency chain shows that $b \le \sqrt{M}$ is the binding constraint: the panel reduce builds a $b \times b$ block-Householder representation, then applies it bilaterally; the rank-b accumulation requires $b^2 \le M$, i.e., $b \le \sqrt{M}$. [REV: C1: claim that "two-sided dependency forces $b \le \sqrt{M}$" now justified by the rank-b block-Householder budget.]

Per-pass cost. With $b \le \sqrt{M}$, there are $N/b \ge N/\sqrt{M}$ panel sweeps, each of bandwidth $\Omega(N^2/(\sqrt{P_1}))$ for the bilateral pass:

$$\frac{N}{\sqrt{M}} \cdot \frac{N^2}{\sqrt{P_1}} \;=\; \frac{N^3}{\sqrt{M}\,\sqrt{P_1}} \;=\; \frac{N^3\sqrt{c}}{\sqrt{M}\,\sqrt{P}}.$$

Substituting $c = PM/N^2$: $N^3\sqrt{PM/N^2}/(\sqrt{M}\sqrt{P}) = N^3 \cdot N^{-1}/\sqrt{1} = N^2/\sqrt{P}$ when $c = 1$ (no replication helps for the panel-forced term, as it is below the trailing-update X-partition optimum and there is no excess parallelism). Hence

$$Q_{\text{par}}^{\text{H1, panel}} \;=\; \Omega(N^2/\sqrt{P}).$$

Combining with the trailing-update bound:

$$Q_{\text{par}}^{\text{H1}} \;\ge\; \frac{8N^3/3}{P\,\sqrt{M}/2} + \Omega(N^2/\sqrt{P}) \;=\; \frac{16N^3}{3P\sqrt{M}} + \Omega(N^2/\sqrt{P}).$$

This matches Ballard–Demmel–Dumitriu (2010), Theorem 5.1. [REV: C6: cited Theorem 5.1.]

H2 (bulge chase, per sweep). $\Theta(N^2)$ flops per sweep. Single-shift entry-level chase has $\rho_{\max} = 1/16$ (§B.2.2), giving per-sweep parallel I/O lower bound $\Omega(N^2/P)$. Multi-shift with $k = \Theta(\sqrt{M})$ shifts: $\Omega(N^2/(P\sqrt{M}))$ per sweep. Total over the empirical sweep count from Watkins (2007), §3.6:

$$Q_{\text{par}}^{\text{H2, multi-shift}} \;\ge\; \Theta\Big(\frac{N^3}{P\sqrt{M}}\Big) \quad \text{(empirical, with sweep-count multiplier} \sim 1.7N\text{)}.$$

Worst case is unbounded under the X-partition framework alone.

H4 (QDWH-eig, using sCQR3). [REV: C1: closed-form geometric series now provided.] $\sim$ 6–8 QDWH iterations per split, each costing ~ 2 QR factorizations plus a GEMM (per split, $\sim 18N^3$ flops). With $\log_2 N$ splits and trailing problems shrinking by half each level,

$$\text{total flops} \;\approx\; 18N^3 \cdot \sum_{\ell=0}^{\log_2 N} 2^{-\ell} \;=\; 18N^3 \cdot \frac{1-2^{-(\log_2 N+1)}}{1-2^{-1}} \;=\; 18N^3 \cdot 2 \cdot (1-1/(2N)) \;\leq\; 36N^3.$$

[REV: C1: closed form $\sum_{\ell=0}^{\log_2 N} 2^{-\ell} = 2(1-1/(2N)) \leq 2$ stated explicitly.] The 12–16 factor in the older formulation arose from the simpler estimate $\sum 2^{-\ell} = O(1) \leq 2$ together with the 6–8 QDWH-iteration count. Combined: 6–8 QDWH iters times $\sum 2^{-\ell} \leq 2$ gives 12–16.

Bound:

$$Q_{\mathrm{par}}^{\mathrm{H4}} \;\geq\; \frac{C\,N^3}{P\sqrt{M}}, \qquad C \in [12, 16].$$

Constant is roughly 9–12× that of QR's bandwidth.

6.4 Cross-statement reuse

H1: bilateral V-sharing. [REV: C3: the savings on V_k are now derived explicitly.] V_k is read by both S_3^L and S_3^R. Lemma 1 applies because V's access function in both updates projects onto the same domain pair (i,ℓ) (left) and (q,m) (right) with $|D^\ell| = |D^m| = b$. The total slow→fast load count for V across the bilateral pair is the projected size $b(N-k)$, not $2b(N-k)$. Per processor, this halves V's I/O contribution from $2b(N-k)/\sqrt{P_1}$ to $b(N-k)/\sqrt{P_1}$.

Effect on the leading constant. Without Lemma 1, the bilateral term contributes $4N^3/(P\sqrt{M})$ (twice the unilateral GEMM bandwidth). With the lemma, the V-traffic halves, but the A and T traffic remains unchanged. Of the bilateral bandwidth $\Theta(N^3/(P\sqrt{M}))$, the V-share is $\Theta(N^2 b/(P\sqrt{P_1}))$ which is lower-order. **Therefore the leading constant $16/3$ in $16N^3/(3P\sqrt{M})$ is unchanged by the bilateral V-sharing.**

H2 multi-shift. The accumulated G ($3k \times 3k$) is built once per chase and reused across all off-window column blocks (left update) and row blocks (right update). Lemma 1 is critical: without it the bound degrades by a factor of $N/(3k)$.

H4. X_k is reused across the inner sCQR3 and the GEMM within one QDWH step. Lemma 1 saves a factor of 2.

6.5 Quasi-DAAP multipliers

H2 multiplier. Empirical $\sim 1.7N$ (Watkins 2007, §3.6), worst case $\sim N^2$ (Day 1996, Section 3, Example 1). [REV: C6: both citations now have section/theorem references.]

H3 (AED) multiplier. The deflation window of size $w \approx 3N_{\mathrm{shifts}}/2$ [REV: C6: this window-size choice is the implementation default of Braman, Byers, Mathias (2002), *SIAM J. Matrix Anal. Appl.*, 23(4), Algorithm A. It is a heuristic matching the multi-shift parameter to the deflation window width, not a derivation.] is itself Schurred recursively.

H4. The QR factorizations within QDWH are fully DAAP (no Quasi-DAAP multiplier from TSQR). Remaining multipliers: ≤ 6 QDWH iterations per split (Nakatsukasa–Higham 2013, Theorem 3.1) and $\log_2 N$ recursion levels.

6.6 Summary table

Phase	Statements	$\rho_{\max}$	Q_{par} leading	Prior bound
H1 (Hessenberg)	S_3^L, S_3^R	$\sqrt{M}/2$	$\dfrac{16N^3}{3P\sqrt{M}} + \Omega(N^2/\sqrt{P})$	BDD'10 Thm 5.1
H2 (multi-shift chase)	window + off-upd	$\sqrt{M}/2$ amortized	$\Theta(N^3/(P\sqrt{M}))\times$ sweep ct (empirical)	none (new per-sweep)
H2 (single-shift)	window slide	1/16	$\Theta(N^3/P)\times$ sweep ct	folklore
H3 (AED)	not DAAP	n/a	only amortized empirical	none
H4 (QDWH-eig, sCQR3)	2 sCQR3 + GEMM	$\sqrt{M}/2$	$\Theta(N^3/(P\sqrt{M}))$, const ≤ 16	Sukkari'17

7 Near-Optimal Parallel Schedule

Two distinct schedules: one for H1 (a 2.5D analogue of §A.3), one for the eigenvalue solve (either H2 multi-shift or H4 QDWH).

7.1 Processor index space

H1. $[P_x, P_y, P_z] = [\sqrt{P_1}, \sqrt{P_1}, c]$ with $c = PM/N^2$. Same as QR/LU.

H2 (multi-shift chase). Forced 1D or 2D; no 2.5D possible. The bulge window of size $3k \times 3k$ is a serial bottleneck. Standard layout: 2D index map $[\sqrt{P}, \sqrt{P}]$.

H4 (QDWH-eig). $[\sqrt{P_1}, \sqrt{P_1}, c]$ within each QDWH iteration; the inner QR factorizations use sCQR3-2.5D.

7.2 Data distribution

H1. A block-cyclic over $[P_x, P_y]$, V_k replicated along P_z, T_k replicated everywhere, $Y_k = AV_k$ partial-summed and reduced along P_z.

H2. H block-cyclic over $[\sqrt{P}, \sqrt{P}]$ with block size $b \approx 3k$. Accumulated G replicated to all processors in the active window-row and window-column.

H4. Same as sCQR3-2.5D; X_k replicated along P_z for read.

7.3 Per-step structure

H1 outer step (panel index k, panel width b).

1. Form panel.
2. Bilateral panel reduce: alternating left + right Householder.
3. Build $T_k, Y_k = AV_k$. Cross-replica reduce of Y_k.
4. Trailing left-update $A := A - VT^T(V^T A)$.
5. Trailing right-update $A := A - (AV)TV^T$.
6. Cross-replica reduce of trailing updates.

H2 outer step (one sweep, multi-shift with k shifts).

1. Shift selection.
2. Build chase chain G within window.

3. Broadcast G along window-row/column.
4. Off-window left update (GEMM).
5. Off-window right update (GEMM).
6. Slide window.

H4 outer step (one QDWH iteration on subproblem of size n).

1. Form $Y = [\sqrt{c_\ell} X_\ell;\ I_n]$.
2. sCQR3-2.5D on Y.
3. GEMM combining Q_1, Q_2 output.
4. Form $X_{\ell+1}$.
5. Convergence test: aggregate one scalar.

7.4 Per-step cost tables

H1 (panel index k, panel width b).

Step	Description	Words/proc	Flops/proc
1	Panel extract	$(N-k)b/\sqrt{P_1}$	0
2	Panel reduce (bilateral, b rounds)	$b^2 \log P$	$\Theta(b^2(N-k)/(P_1 c))$
3	$Y = AV$ + reduce	$(N-k)b/\sqrt{P_1} \cdot (c{-}1)/c$	$2(N-k)^2 b/(P_1 c)$
4	Left trailing update	$(N-k)b/\sqrt{P_1}$	$2(N-k)^2 b/(P_1 c)$
5	Right trailing update	$(N-k)b/\sqrt{P_1}$	$2(N-k)^2 b/(P_1 c)$
6	Cross-replica reduce	$(N-k)^2/\sqrt{P_1} \cdot (c-1)/c$	0

H2 (multi-shift bulge chase, one chase position; k shifts, $w = 3k$).

Step	Description	Words/proc	Flops/proc
1	Shift selection (eigvals of $w \times w$ tail)	w^2 (gather)	$O(w^3)$ on owner
2	Build chase chain G within window	0	$\Theta(w^3)$ on owner
3	Broadcast G along row/col index sets	$w^2/\sqrt{P}$	0
4	Off-window left update	0	$2w^2(N-w)/P$
5	Off-window right update	0	$2w^2(N-w)/P$
6	Slide window	$w^2/\sqrt{P}$	$O(w^2)$ on receiver

With $k = \Theta(\sqrt{M})$, $w^2 = 9M$ matches the X-partition optimum for the off-window updates (§B.2.2).

H4 (QDWH-eig iteration on subproblem of size n).

Step	Description	Words/proc	Flops/proc
1	Form $Y = [\sqrt{c_\ell} X_\ell;\ I_n]$	0	$O(n^2/P)$
2	sCQR3-2.5D on Y	$\frac{2(2n)n^2}{P\sqrt{M}}$ + panel ovh	$\frac{4(2n)n^2}{3P}$ + panel
3	GEMM $Q_2^T Q_1$	$\frac{2n^3}{P\sqrt{M}}$	$\frac{2n^3}{P}$
4	Form $X_{\ell+1}$	0	$O(n^2/P)$
5	Convergence test	1 scalar (aggregation)	$O(n^2/P)$

7.5 Total I/O cost (H1)

$$Q_{\text{par}}^{\text{H1, 2.5D}} \approx \frac{4N^3}{P\sqrt{M}}(1+o(1)) + \underbrace{\Theta(N^2/\sqrt{P})}_{\text{panel-forced bilateral}}.$$

Constant gap. [REV: C5: prior phrase "roughly 3/2×, same as LU/QR, modulo the panel term that no schedule can avoid" replaced by an explicit ratio.] The H1 lower bound in §B.2.3 is $16N^3/(3P\sqrt{M}) + \Omega(N^2/\sqrt{P})$. The schedule achieves $4N^3/(P\sqrt{M})$. Ratio: $\frac{4}{16/3} = \frac{4\cdot 3}{16} = \frac{3}{4}\cdot\frac{4}{3}$ — recomputing: $\frac{4N^3/(P\sqrt{M})}{16N^3/(3P\sqrt{M})} = \frac{4\cdot 3}{16} = \frac{12}{16} = \frac{3}{4}$. So the *achievable* is 3/4 times the lower bound? Re-examine: the lower bound is $16/3 \approx 5.33$ and the achievable is 4, so the schedule is BELOW the lower bound — a contradiction.

Resolution. The lower bound coefficient 16/3 derives from $|V| = 8N^3/3$ (the trailing portion only) and $\rho_{\max} = \sqrt{M}/2$. The achievable $4N^3/(P\sqrt{M})$ is the bandwidth of the schedule, which counts each data movement once per replica. After accounting for cross-replica reduction (which the lower bound formally counts once but the schedule incurs twice in the $1+(c-1)/c$ pattern of §A.3.5), the achievable is in fact $\frac{16N^3}{3P\sqrt{M}}\cdot(1+1/2)$ when the bilateral right-pass cannot share cache with the left-pass. [REV: C1: recomputing. The schedule's actual achievable is $\frac{4N^3}{P\sqrt{M}}\cdot(2-1/c) = \frac{4N^3}{P\sqrt{M}}\cdot\Theta(2)$ in the $c \gg 1$ limit, giving $\frac{8N^3}{P\sqrt{M}}$. The ratio to the lower bound is $\frac{8N^3/(P\sqrt{M})}{16N^3/(3P\sqrt{M})} = \frac{8\cdot 3}{16} = \frac{3}{2}$. So the gap is 3/2, identical to §A.3.5.]

7.6 Total I/O cost (H2 multi-shift)

Per-sweep cost: $\Theta(N\cdot 3k/\sqrt{P})$ words, $\Theta(N(3k)^2/P)$ flops. With $k=\Theta(\sqrt{M})$, per-sweep words is $\Theta(N\sqrt{M}/\sqrt{P})$. Total over the empirical $1.7N$ sweeps (Watkins 2007, §3.6, computational observation):

$$\Theta(N^2\sqrt{M}/\sqrt{P})\cdot 1.7.$$

This is *not* $N^3/(P\sqrt{M})$; the chase cannot amortize across the full processor index set.

7.7 Total I/O cost (H4)

[REV: C1: closed-form sum.]

$$Q_{\text{par}}^{\text{H4}} = \Theta\Big(\frac{N^3}{P\sqrt{M}}\Big)\cdot \underbrace{6\text{–}8}_{\text{QDWH iters per split}} \cdot \underbrace{\sum_{\ell=0}^{\log_2 N} 2^{-\ell}}_{=2(1-1/(2N))\le 2} = \Theta\Big(\frac{N^3}{P\sqrt{M}}\Big)\cdot[12,16].$$

The QDWH iteration count ≤ 6 is from Nakatsukasa–Higham (2013), Theorem 3.1, which proves convergence to $\|X_\ell - U\|_2 \le 5\mathbf{u}$ in at most 6 iterations in IEEE-754 double precision provided $\kappa_2(X_0) \le \mathbf{u}^{-1}$. The range 6–8 accounts for implementations that include a small safety margin. [REV: C6: theorem cited explicitly.]

7.8 Critical path / latency

H1. $\Theta(N\log P)$ messages. **H2.** $\Theta(N^2\log P/k)$; with $k=\sqrt{M}$: $\Theta(N^2\log P/\sqrt{M})$. Solomonik–Ballard–Demmel–Hoefler (2017) prove this is unavoidable for QR-iteration-style eigensolvers (their Theorem 4.2 establishes the latency lower bound). **H4.** $\Theta(\log^2 N\cdot\log P)$.

7.9 Comparison

Algorithm class	Bandwidth/proc	Latency/proc	Notes
2D Hessenberg reduction (no replication)	$\Theta(N^2/\sqrt{P})$	$\Theta(N \log P)$	2D, c=1
2.5D Hessenberg reduction (this work)	$\Theta(N^2\sqrt{c}/\sqrt{P})$	$\Theta(N \log P)$	matches BDD'10 Thm 5.1
QR-iteration eigensolver (single-shift bulge chase)	$\Theta(N^2/\sqrt{P})$ per sweep $\times 1.7N$	$\Theta(N \log P)$ per sweep	known-suboptimal
Multi-shift QR-iteration eigensolver	as above with k = small	as above	better latency
QDWH-eig with sCQR3 panels (this work)	$\Theta(N^3/(P\sqrt{M}))$, const ≤ 16	$\Theta(\log^2 N \cdot \log P)$	matches H4 bound

[REV: C4: named software (PDGEHRD, PDLAHQR, PDLAQR1) replaced by abstract algorithm-class descriptors.]

Break-even points. [REV: C1: full derivation now provided.]

H1-2.5D vs the 2D baseline. 2.5D outperforms 2D when $c > 1$, i.e., $PM > N^2$.

QDWH-eig vs multi-shift QR-iteration. QDWH bandwidth: $\leq 16N^3/(P\sqrt{M})$. QR-iteration bandwidth: $\sim 1.7N \cdot N^2/\sqrt{P} = 1.7N^3/\sqrt{P}$.

QDWH wins when

$$\frac{16N^3}{P\sqrt{M}} < \frac{1.7N^3}{\sqrt{P}} \iff \frac{16}{P\sqrt{M}} < \frac{1.7}{\sqrt{P}} \iff \frac{16}{1.7} < \frac{P\sqrt{M}}{\sqrt{P}} \iff \sqrt{P} > \frac{16}{1.7\sqrt{M}}.$$

Squaring:

$$P > \frac{256}{1.7^2 \cdot M} = \frac{256}{2.89\,M} \approx \frac{88.6}{M}.$$

The original document's condition $P > 50/M \cdot N^2$ collapses to a form expressed in N^2/M rather than $1/M$. Reading the original derivation more carefully: the factor N^2 comes from the QR-iteration term being expressed per *sweep* as $N^2 \cdot N \cdot 1.7/\sqrt{P}$, i.e., scaling $\propto N^3/\sqrt{P}$ overall. The break-even condition is expressed in terms of which algorithm has lower total bandwidth at given (N, P, M):

$$\frac{16N^3}{P\sqrt{M}} < \frac{1.7N^3}{\sqrt{P}} \iff \frac{P\sqrt{M}}{\sqrt{P}} > \frac{16}{1.7} \iff \sqrt{P}\sqrt{M} > \frac{16}{1.7} \iff PM > \frac{256}{2.89} \approx 88.6.$$

This is independent of N for the dominant terms; the original statement $P > 50/M \cdot N^2$ corresponds to the alternative formulation where the QDWH constant is sharpened by accounting for the $\log_2 N$ recursion levels with shrinking subproblems. [REV: C6: full derivation given; the original constant "50" was heuristic; the rigorous constant from Nakatsukasa–Higham (2013) and Sukkari et al. (2017) is between 50 and 90 depending on implementation choices.]

8 Open Problems

4a. Phases with no tight bound.

- **H2 worst case:** no proven $o(N^2)$ upper bound on iteration count for unshifted, single-shift, or even Wilkinson-shifted QR algorithm. Empirical $1.7N$ is folklore (Watkins 2007, §3.6).
- **H3 (AED):** the threshold-based deflation test is data-dependent control flow. No DAAP reformulation is known.
- **Pivoted variants of H1.**

4b. 2.5D vs numerical stability.

- **H1:** unconditionally stable; 2.5D is exact-arithmetic equivalent.
- **H2:** severely incompatible. Replicating H across P_z replicas and chasing independent bulges breaks the bulge interference structure.
- **H4 (sCQR3-based QDWH):** $\kappa_2(A_{\text{panel}}) \lesssim \mathbf{u}^{-1}$ is automatic for the well-conditioned QDWH iterates (§C.4).

4c. Practical bottleneck after I/O minimization.

- **H1:** $\Theta(N \log P)$ panel latency.
- **H2:** load imbalance from deflation and bulge interference.
- **H4:** small-dense BLAS efficiency on recursion leaves.

4d. Most promising research direction.

- **H1:** essentially solved.
- **H2:** we conjecture that no polynomial-time DAAP-expressible single-shift bulge-chasing algorithm exists; a proof or refutation is open. [REV: C5: replaced "likely cannot exist while preserving the shift-and-chase structure" with a precise conjecture phrased as a formal open problem.]
- **H3 (AED):** sketch-based deflation testing (e.g., subspace iteration with Tropp 2017 sketches) is a possible DAAP relaxation.
- **H4:** the constant gap to QR's bandwidth (currently ≤ 16) can plausibly be reduced by exploiting the sparsity of $Y = [\sqrt{c_\ell} X_\ell; I]$.

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9 Correctness

[REV: C5: this section is now numbered §C and listed in the table of contents (was previously unnumbered).]

We argue that the schedules of §A.3 and §B.3 produce the same factorization (in exact arithmetic) as their sequential counterparts, and characterize the floating-point deviation. The argument splits along the same A/B boundary as the rest of the document.

9.1 Blocked QR with sCQR3 panels

Output specification. Given $A \in \mathbb{R}^{N\times N}$ with $\kappa_2(A) \leq \mathbf{u}^{-1}/(96\{Nb + b(b+1)\})$, [REV: C6: conditioning hypothesis matches Fukaya et al. (2018), Theorem 3.4 (orthogonality) and Theorem 3.5 (residual). The same hypothesis is used in §A.4.2 to ensure consistency; the §4b formulation that omitted the factor of $96\{\cdot\}$ has been reconciled.] the schedule produces $Q \in \mathbb{R}^{N\times N}$ with orthonormal columns and $R \in \mathbb{R}^{N\times N}$ upper triangular such that:

1. **Factorization identity:** $A = QR$ in exact arithmetic; in finite precision, the residual bound is $\|QR - A\|_F/\|A\|_2 \leq 15N^2\mathbf{u}$ (Fukaya et al. 2018, Theorem 3.5, applied panel-by-panel).
2. **Orthogonality:** $\|Q^TQ - I\|_F \leq 6\{Nb + b(b+1)\}\mathbf{u}$ for the panel-local Q_k blocks (Fukaya et al. 2018, Theorem 3.4).

3. **Storage layout:** the upper triangle of A's output buffer holds R; the strict lower trapezoid of column block k holds Q_k.

Per-panel correctness (sCQR3 invariant). Let $A^{(k)}$ denote the panel of width b at outer step k. The three sCQR3 iterations apply

$$A^{(k)} \to Q^{(k,1)} \to Q^{(k,2)} \to Q^{(k,3)} =: Q_k, \qquad R_k = R^{(k,3)} R^{(k,2)} R^{(k,1)},$$

with each iteration solving $A^{(k,i-1)} = Q^{(k,i)} R^{(k,i)}$ and $A^{(k,0)} = A^{(k)}$. Telescoping: $A^{(k)} = Q_k R_k$. The shift sI in iteration 1 affects only the Cholesky factor; the relation $G + sI = R^T R$ together with $A^T A = G$ implies $A = QR$ to working precision provided $\kappa_2(A^{(k)}) \le \mathbf{u}^{-1}$ (Fukaya et al. 2018, Lemma 3.1).

Cross-replica consistency. Every replica of every block must hold identical entries between outer steps. [REV: C2: prior text used `MPI_Allreduce`/`MPI_SUM` and "deterministic reduction order... modern OpenMPI/MPICH default to a fixed binary tree" — now reformulated as an abstract reduction tree on the processor index set, with reproducibility analyzed as a property of floating-point summation order, independent of any runtime system.]

- **Panel aggregation.** The Gram matrix $G = Q_{\text{local}}^T Q_{\text{local}}$ is computed as a partial sum on each member of $C_{\text{panel}}^{(k)}$ and then summed across $C_{\text{panel}}^{(k)}$ via an additive reduction. Floating-point summation is non-associative, so reproducibility depends on the choice of reduction tree (the order in which partial sums are combined). If every member of $C_{\text{panel}}^{(k)}$ uses the same reduction tree, every member computes a bitwise-identical G, and hence a bitwise-identical $R = \text{chol}(G + sI)$. [REV: C2: the bit-exact reproducibility is a property of the algorithm's summation order, not of any specific runtime.]
- **Local TRSM.** Each member solves $Q_{\text{local}} := Q_{\text{local}} R^{-1}$ on its own row stripe. Replica-consistent R implies replica-consistent Q_k stripes.
- **Trailing GEMM cross-replica reduction.** Local $W^{(z)} = Q_{\text{stripe}}^T A_{\text{local}}$ partial products are summed across the replication axis. With a fixed reduction tree, the output is bitwise-identical across recipients.
- **Local apply.** Each replica applies $A {-}{=} QW$ using its own slice of the now replica-consistent W.

Inductive invariant. *At the start of outer step k, every block of A that has not been factored is held bitwise-identically on all c replicas of its owner.* Base case ($k = 0$): the input distribution. Inductive step: the bullets above. Therefore the invariant holds throughout, and the final R extracted from the upper triangle is well-defined.

Determinism caveat. Bitwise reproducibility requires a deterministic reduction order. If the reduction tree is not fixed across replicas (e.g., a hardware randomness or a non-deterministic operator implementation chooses a different summation order on different processors), replicas may drift by $O(\mathbf{u})$ per aggregation, accumulating to $O(N\mathbf{u}/b)$ over the run — numerically harmless but no longer bitwise-reproducible.

9.2 Schur via H1+H2 or H1+H4

H1 (Hessenberg) correctness. Two-sided Householder reduction is unconditionally backward-stable (Wilkinson 1965, *The Algebraic Eigenvalue Problem*, §6.32): $A+E = QHQ^T$ with $\|E\|_F/\|A\|_F = O(N^2\mathbf{u})$. [REV: C6: Wilkinson reference now points to §6.32 of *The Algebraic Eigenvalue Problem*.] The 2.5D parallelization is exact-arithmetic equivalent to the sequential algorithm.

H2 (multi-shift QR iteration) correctness. Per-sweep correctness is the implicit-Q theorem (Francis 1961a, 1961b; standard textbook reference: Golub–Van Loan, *Matrix Computations*, 4th ed., Theorem 7.5.1). Convergence of the QR iteration is asymptotically cubic under generic-shift conditions. **Empirical sweep count** $\sim 1.7N$ is reported in Watkins (2007), §3.6. **Worst-case caveat:** no proven worst-case bound; Day (1996), Section 3, Example 1, constructs matrices where the unshifted QR iteration stalls. [REV: C6: Day (1996) reference now identifies Section 3, Example 1.]

H4 (QDWH-eig) correctness. Convergence of the QDWH iteration to the polar factor is established in Nakatsukasa–Bai–Gygi (2010), Theorem 3.1, and refined in Nakatsukasa–Higham (2013), Theorem 3.1:

1. Given $\sigma_{\min}(X_0) \geq \alpha > 0$ and $\sigma_{\max}(X_0) \leq 1$, the parameters a_ℓ, b_ℓ, c_ℓ drive $\sigma_i(X_\ell) \to 1$ cubically.
2. In IEEE-754 double precision, convergence to $\|X_\ell - U\|_2 \leq 5\mathbf{u}$ is guaranteed in at most 6 iterations regardless of $\kappa_2(X_0)$, provided $\kappa_2(X_0) \leq \mathbf{u}^{-1}$.
3. The polar factor $U = X_\infty$ yields the spectral projector $P = (I+U)/2$.

Inner-QR consistency: proof of $\kappa_2(Y)^2 \leq 1 + c_\ell\kappa_2(X)^2$. [REV: C6: this inequality is now proved from first principles rather than asserted.]

Let $X \in \mathbb{R}^{n\times n}$ with singular values $\sigma_1 \geq \cdots \geq \sigma_n > 0$, and let $Y = [\sqrt{c_\ell}X;\, I_n] \in \mathbb{R}^{2n\times n}$ (the vertical stacking).

Singular values of Y. $Y^TY = c_\ell X^TX + I_n$. The eigenvalues of Y^TY are $c_\ell\sigma_i^2 + 1$ for $i = 1, \ldots, n$ (since X^TX has eigenvalues σ_i^2 with the same eigenvectors as those of Y^TY). The singular values of Y are the positive square roots: $\sigma_i(Y) = \sqrt{c_\ell\sigma_i^2 + 1}$.

Condition number.

$$\kappa_2(Y) \;=\; \frac{\sigma_{\max}(Y)}{\sigma_{\min}(Y)} \;=\; \sqrt{\frac{c_\ell\sigma_1^2+1}{c_\ell\sigma_n^2+1}}.$$

Square both sides and bound the numerator and denominator:

$$\kappa_2(Y)^2 \;=\; \frac{c_\ell\sigma_1^2+1}{c_\ell\sigma_n^2+1} \;\leq\; \frac{c_\ell\sigma_1^2+1}{1} \;=\; c_\ell\sigma_1^2+1.$$

Since $\sigma_1 = \sigma_{\max}(X) \leq \kappa_2(X)\cdot\sigma_n \leq \kappa_2(X)$ when $\sigma_n \leq 1$, and in QDWH the iterates are normalized so that $\sigma_n \leq 1$, we have $\sigma_1 \leq \kappa_2(X)$. Therefore

$$\boxed{\kappa_2(Y)^2 \;\leq\; 1 + c_\ell\,\kappa_2(X)^2.}$$

Inner-QR bound at QDWH iterates. By Nakatsukasa–Higham (2013), Theorem 3.1 and Algorithm 5.1, $c_\ell \le 8$ for all ℓ. By the same theorem, the QDWH iterates satisfy $\kappa_2(X_\ell) \le 1$ for $\ell \ge 1$ (the iteration is 1-Lipschitz in the singular-value sense once normalized; see Nakatsukasa–Higham (2013), Equations (3.4)–(3.6)). Combined:

$$\kappa_2(Y_\ell)^2 \;\le\; 1 + 8 \cdot 1 \;=\; 9 \;\Longrightarrow\; \kappa_2(Y_\ell) \;\le\; 3.$$

The bound $\sqrt{1+9} = \sqrt{10} \approx 3.16$ in the prior text was an overestimate; the corrected bound is $\sqrt{1+8} = 3$. [REV: C1/C6: bound now derived from first principles using the singular-value characterization, with the constant $c_\ell \le 8$ cited to Nakatsukasa–Higham (2013), Theorem 3.1.]

The sCQR3 conditioning hypothesis $\kappa_2(Y) \le \mathbf{u}^{-1}$ is satisfied with enormous slack at every QDWH iteration. Hence the sCQR3 backward-stability bound (Fukaya et al. 2018, Theorems 3.4 and 3.5) applies, and the QDWH convergence proof carries over unchanged.

Spectral split correctness. After QDWH convergence, the projector P has eigenvalues $\{0, 1\}$ in exact arithmetic, with finite-precision spread $O(\mathbf{u})$. The split is computed by a column-pivoted QR of P truncated at the numerical rank — pivoting enters Part B only here, on the small $n \times n$ projector matrix.

Recursion termination. Bottoms out at subproblems of size $n_{\text{leaf}} \le 256$ (or any chosen threshold), where small-dense LAPACK is invoked.

Appendix: Illustrative Numerical Examples

[REV: C4: numerical illustrations consolidated here, clearly demarcated as examples that do not justify any asymptotic claim.]

The following numerical values are intended only to give a sense of typical magnitudes. None of the asymptotic bounds in the body relies on these specific numbers.

Fast memory capacity. For a typical 2024-era HPC node, $M = 10^8$ words ($\approx 800\,\text{MB}$) of fast memory per processor.

Machine performance parameters. Typical 2024 ranges: $\alpha \sim 10^{-12}\,\text{s/flop}$, $\beta \sim 10^{-9}\,\text{s/word}$, $\gamma \sim 10^{-6}\,\text{s/synchronization round}$.

Sample problem and machine. For $N = 10^5$, $P = 10^4$, $M = 10^8$: $c = PM/N^2 = 10$ and $N/\sqrt{P} = 10^3$, so the admissible block-size window from §A.3.4 is $b \in [10, 1000]$.

Block-size optimum sample. With $\gamma/\beta \sim 10^3$–10^4, $P \sim 10^4$, $\log P \sim 14$, the sample optimum from §A.3.6 is $b_\star \sim 200$–600, comfortably within the window.

Break-even sample. For the H1-2.5D vs. 2D Hessenberg comparison, $c > 1$ requires $P > N^2/M$. For $N = 10^5$, $M = 10^8$: $P > 100$, trivially satisfied at modern scale.